

Bootstrapping AI Adoption for Smart Manufacturing

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Chief Scientist of Machine Learning



自主創新
Rapid Innovation



綠能環保
Sustainable Energy



雲端應用
Cloud Solutions



移動生活
Mobile Lifestyle



新興市場
Emerging Markets



Carnegie Mellon University (CMU) Ph.D.



1st Video Processor

Nvidia Video Architect



14 million users

Intel OpenCV Many-core



More queries than Google

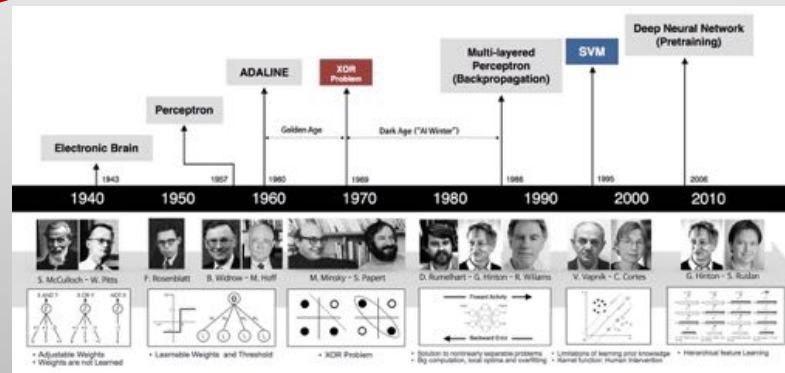
CV startup founder + Team of acquired startup



Inventec AI Center Chief Scientist of ML



1995
SVM birth



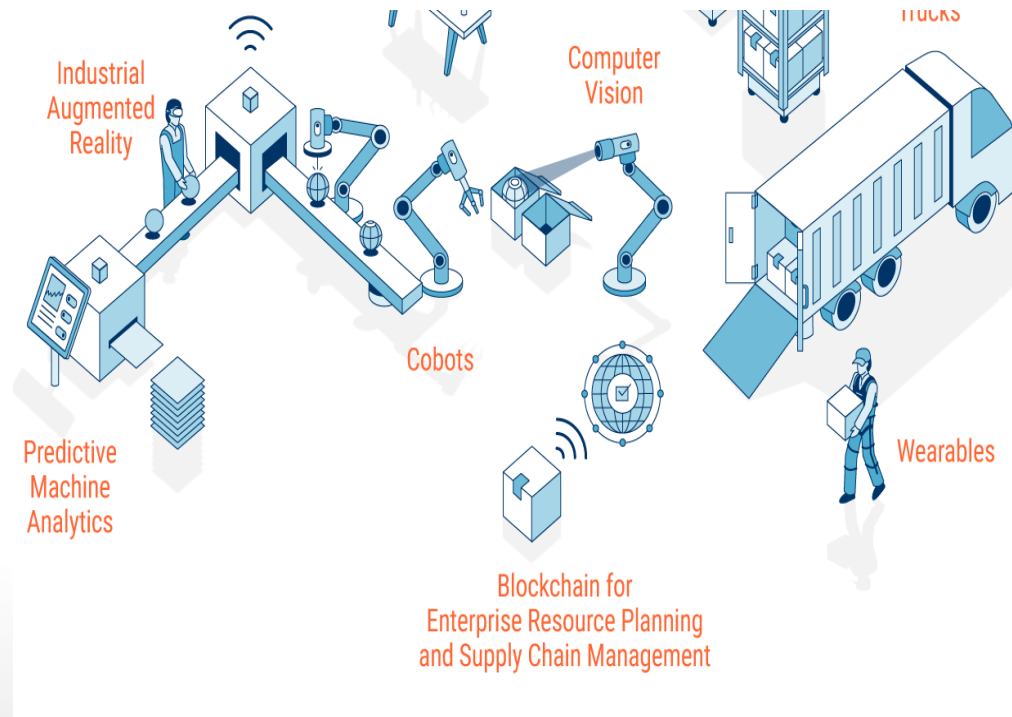
2018
AI for Manufacturing

What you think AI does...



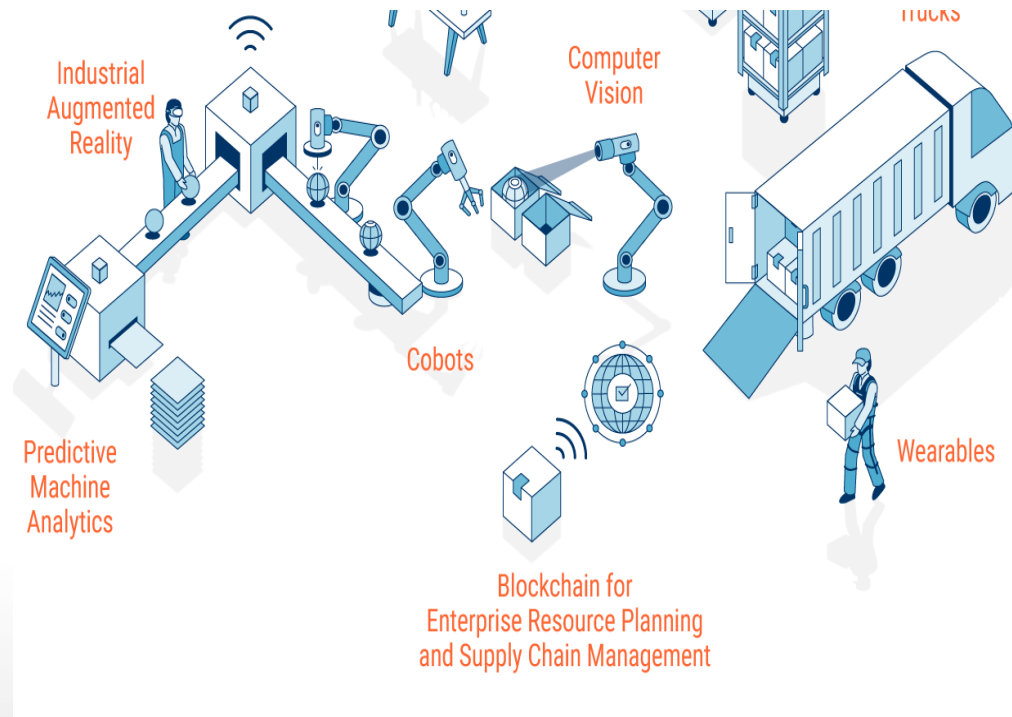
What it really does...





AI • Manufacturing

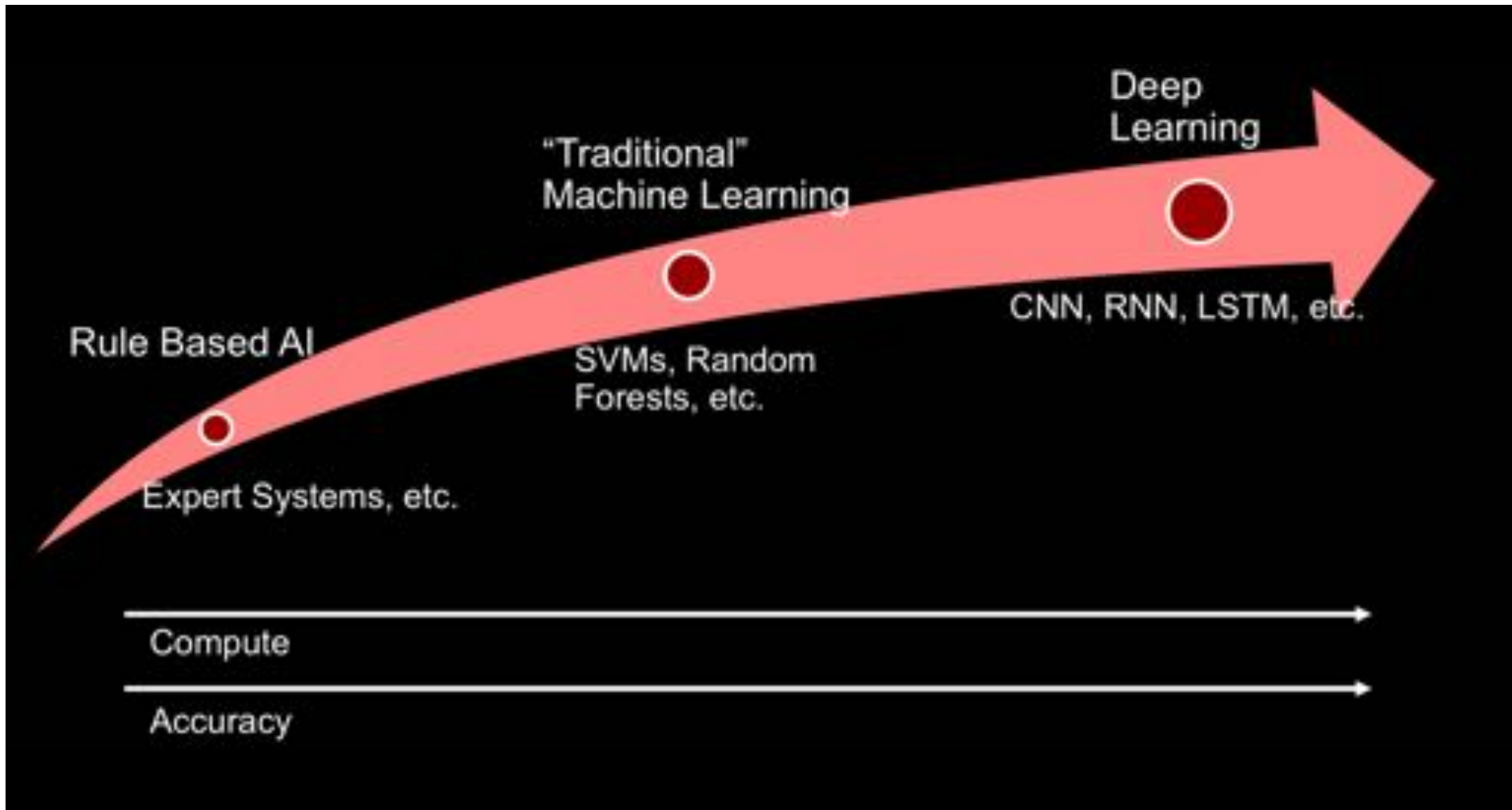
Adopting AI for Manufacturing



AI • Manufacturing

Adopting AI for Manufacturing

Artificial Intelligence



DEEP LEARNING

TRAINING

Learning a new capability
from existing data

Untrained
Neural Network
Model



Deep Learning
Framework

TRAINING
DATASET



Trained Model
New Capability



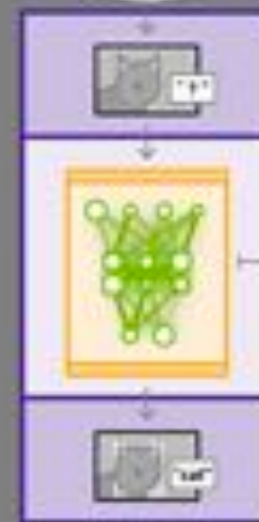
INFERENCE

Applying this capability
to new data

NEW
DATA



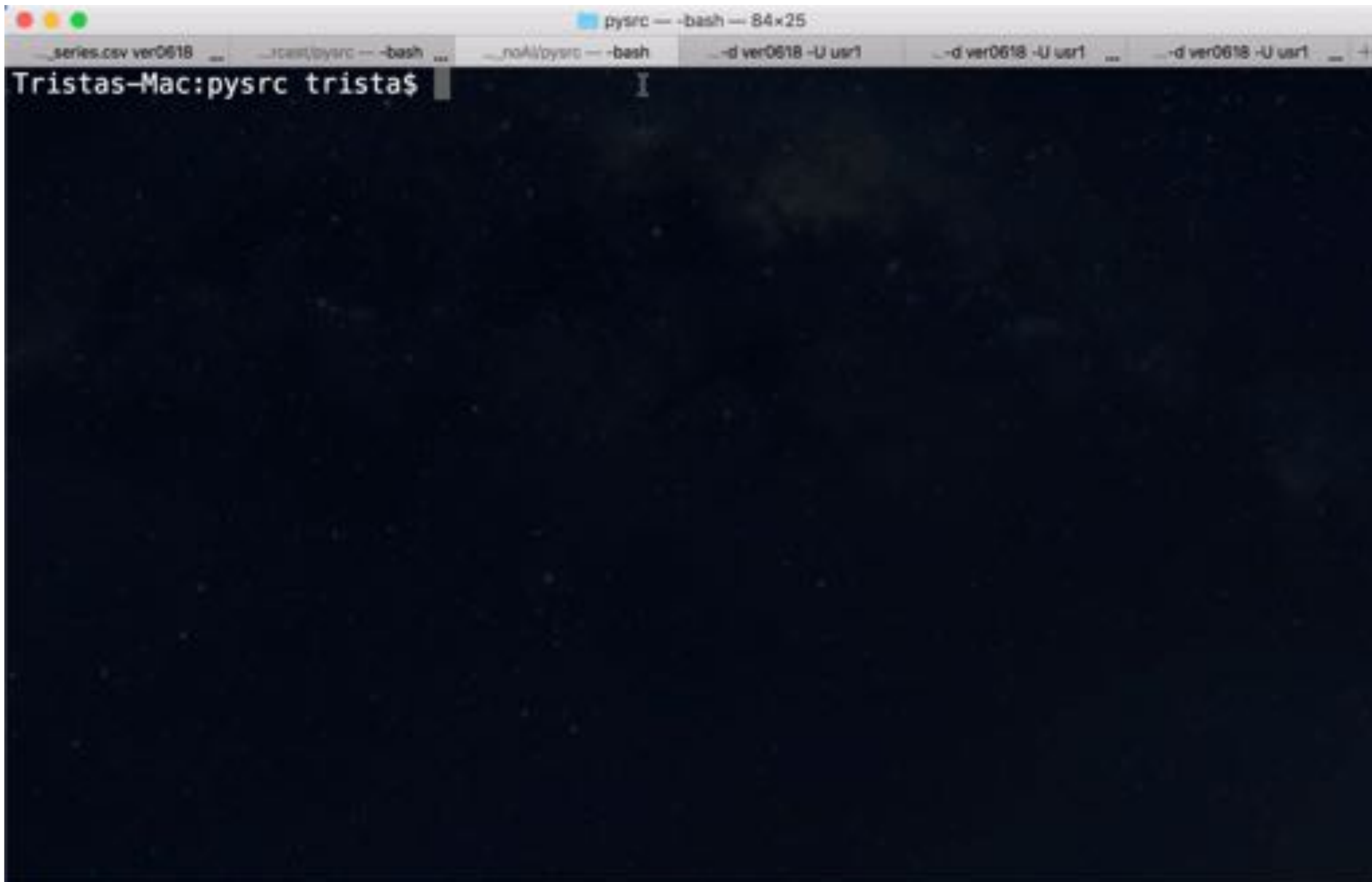
App or Service
Featuring Capability

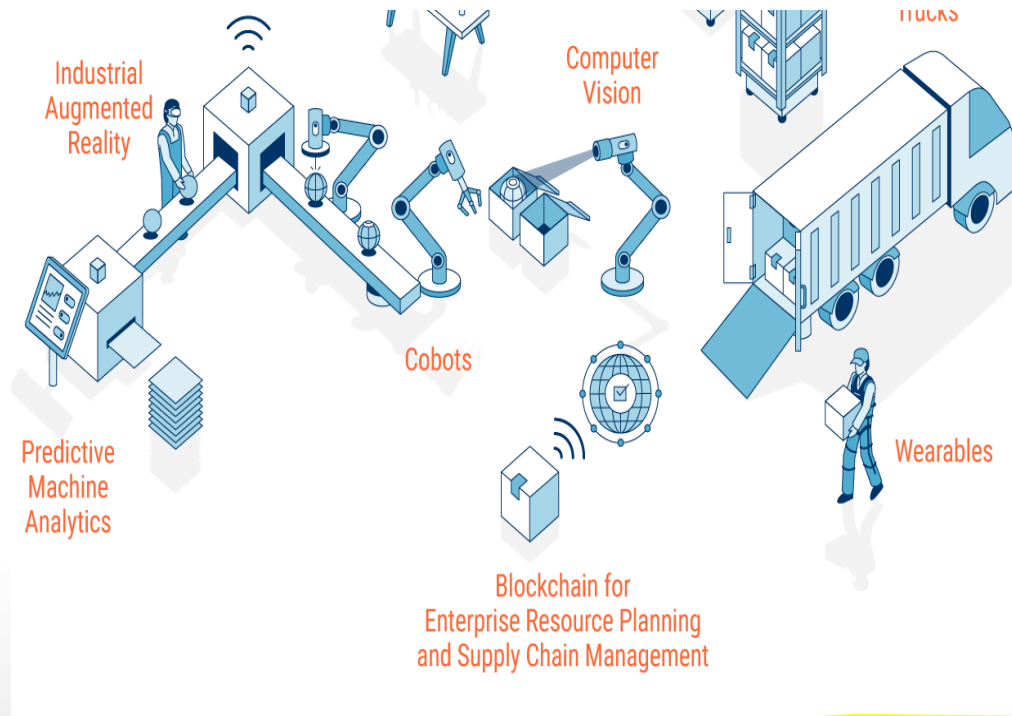


Trained Model
Optimized for
Performance

Deep Learning can do EVERYTHING ?!

$$\sqrt{x_1} + \sqrt{x_2} + \sqrt{x_3} + \sqrt{x_4} = ?$$





AI • **Manufacturing**

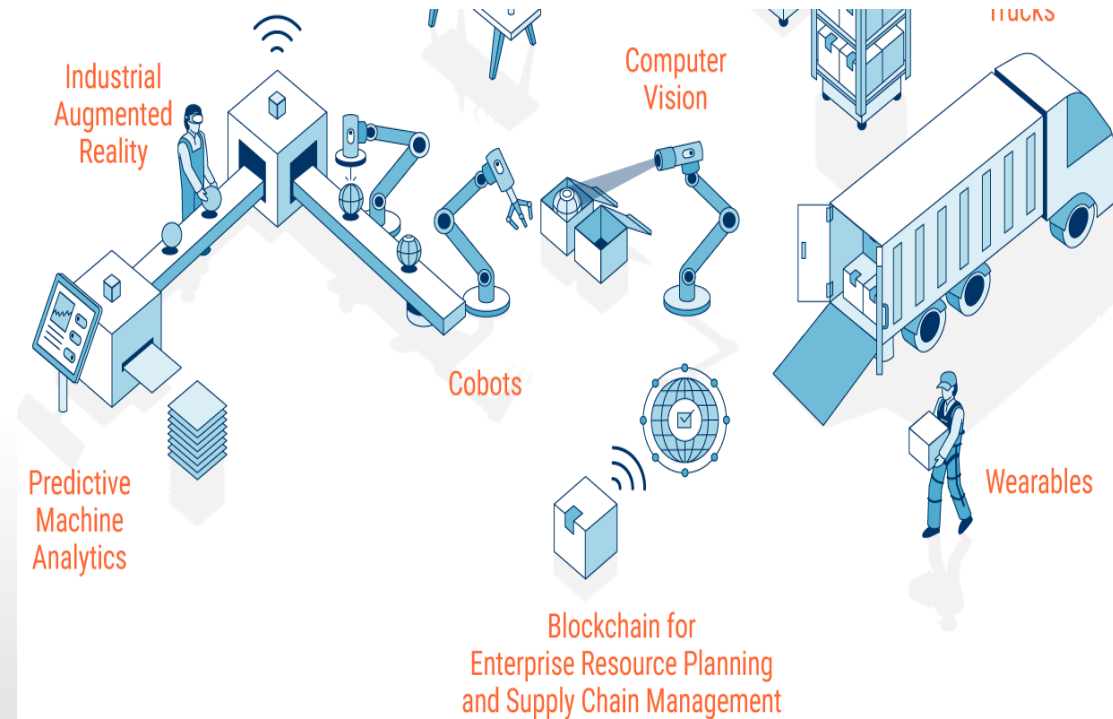
Adopting AI for Manufacturing

Manufacturing stores more data than any other sector
– an estimated two exabytes ($2 \cdot 10^{18}$ bytes) in 2010.

Source: McKinsey & Company, *"Is manufacturing 'cool' again?"* 2013

AI for Smart Manufacturing

- Optical inspection (AOI) for defect detection
- AI/computer vision for automation
- Facility/job scheduling
- Preventive maintenance
- Production prediction: yield, chip area, etc.
- Business decisions: order placements, resource allocations and managements, etc.



AI for Smart Manufacturing: Case studies

Real-world cases at Inventec:

1. AOI re-inspection for SMT lines and PCBs
2. Automatic inspection of laptop surface defects
3. Latte: LApTop TEsting Robot

Inventec at a Glance

- Public company founded 1975 in Taipei, Taiwan.
- Annual revenue USD15+ billions (2017)
- Top laptop, cloud server, and smart device manufacturer in the world

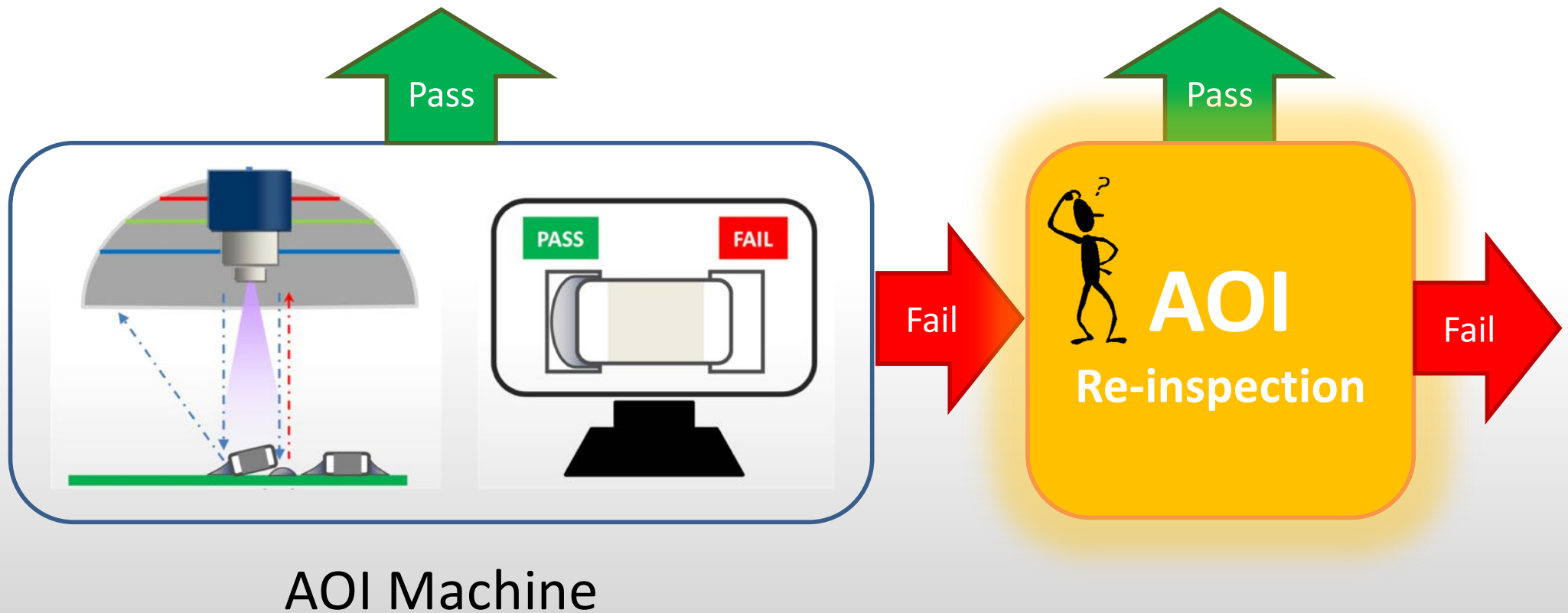


AI for Smart Manufacturing: Case studies

Real-world cases at Inventec:

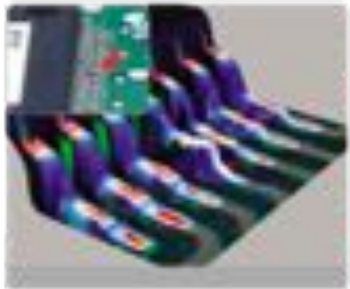
1. AOI re-inspection for SMT lines and PCBs
2. Automatic inspection of laptop surface defects
3. Latte: LApTop TEsting Robot

1: AOI re-inspection: problem definition



By: 許正翰, 呂明峻, 張文于 of NTU & advised by Wei-chao Chen of Skywatch & NTU & Trista Chen of Inventec

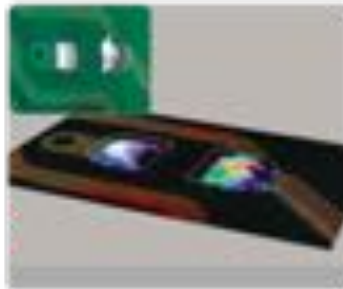
1: AOI re-inspection: problem definition



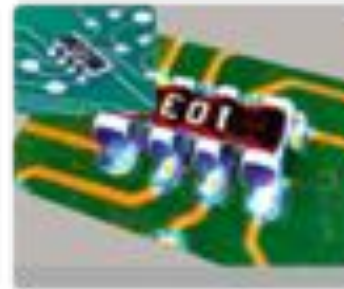
Lead Lift



Component Lift



Missing



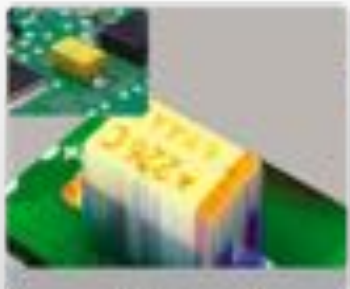
Misalignment



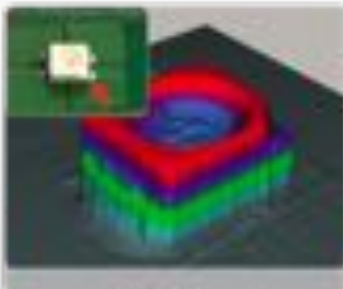
Tombstoning



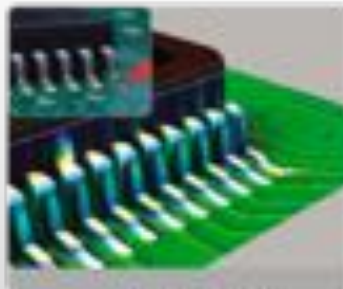
Upside Down



POLARITY



POLARITY



Solder Joint



Bridge



Text

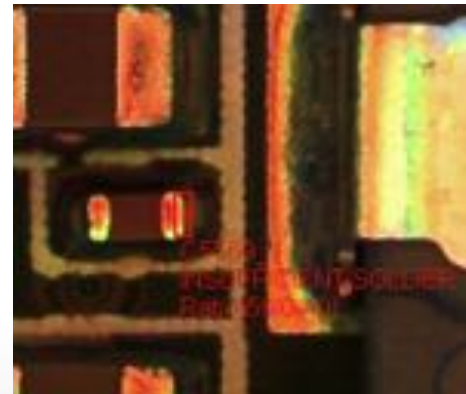
1: AOI re-inspection: data

Insufficient solder

FAIL



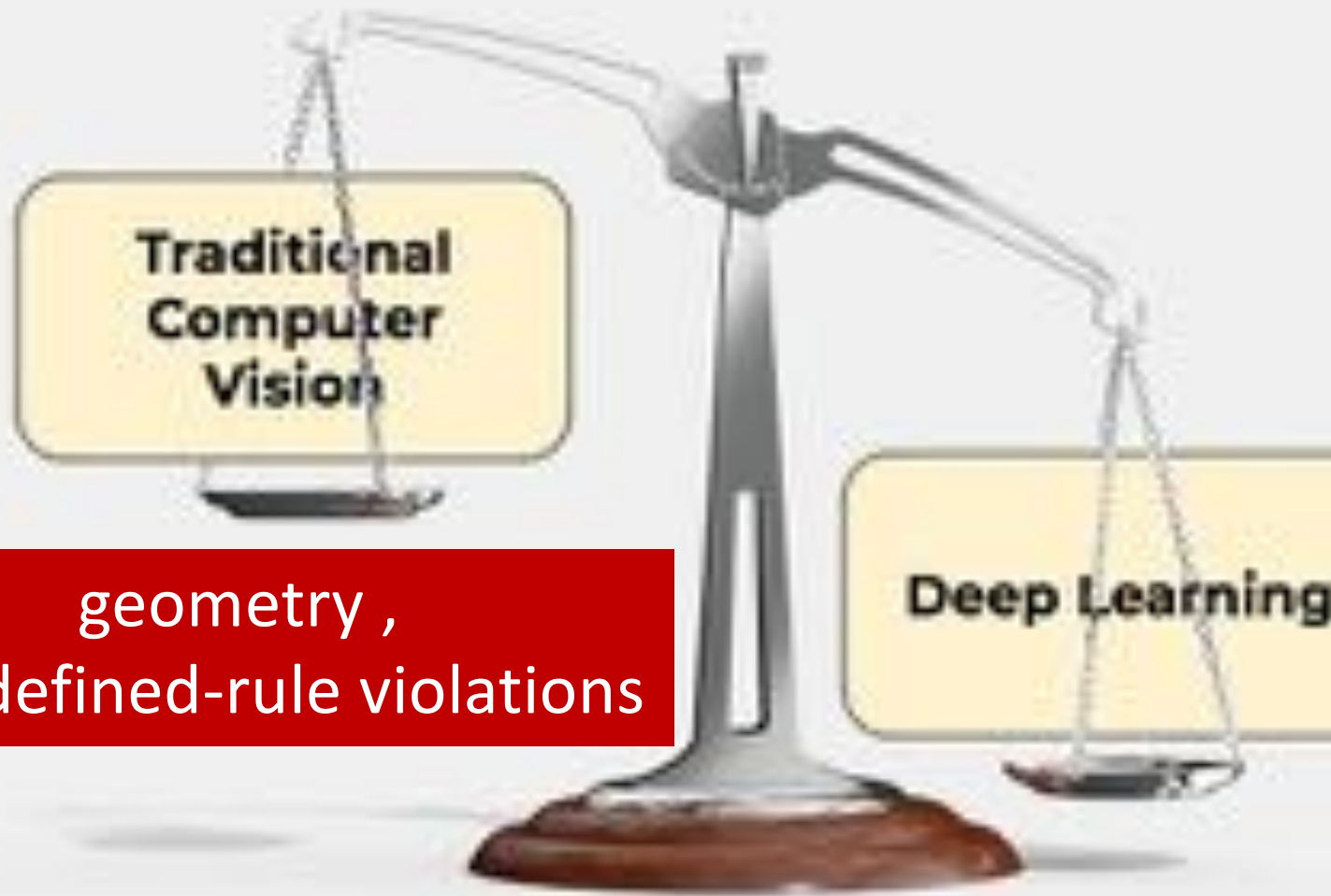
PASS



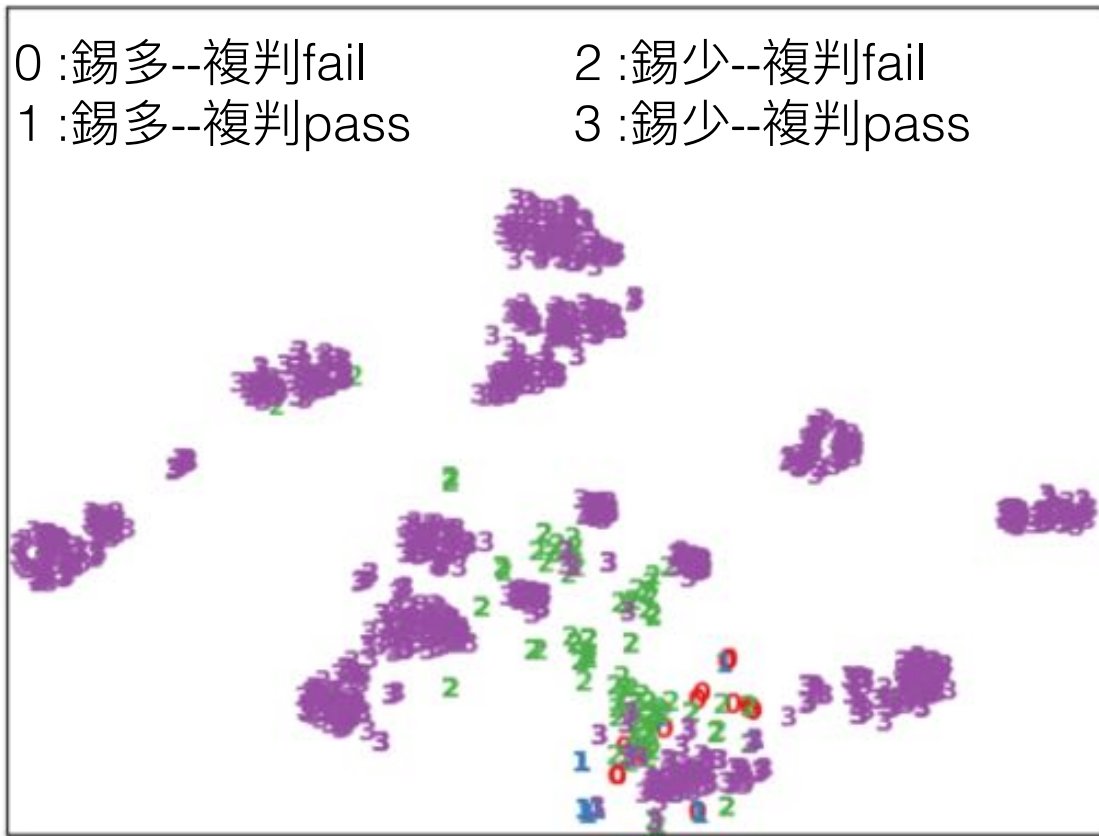
Overflow solder



1: AOI re-inspection: model selection



1: AOI re-inspection: result in accuracy



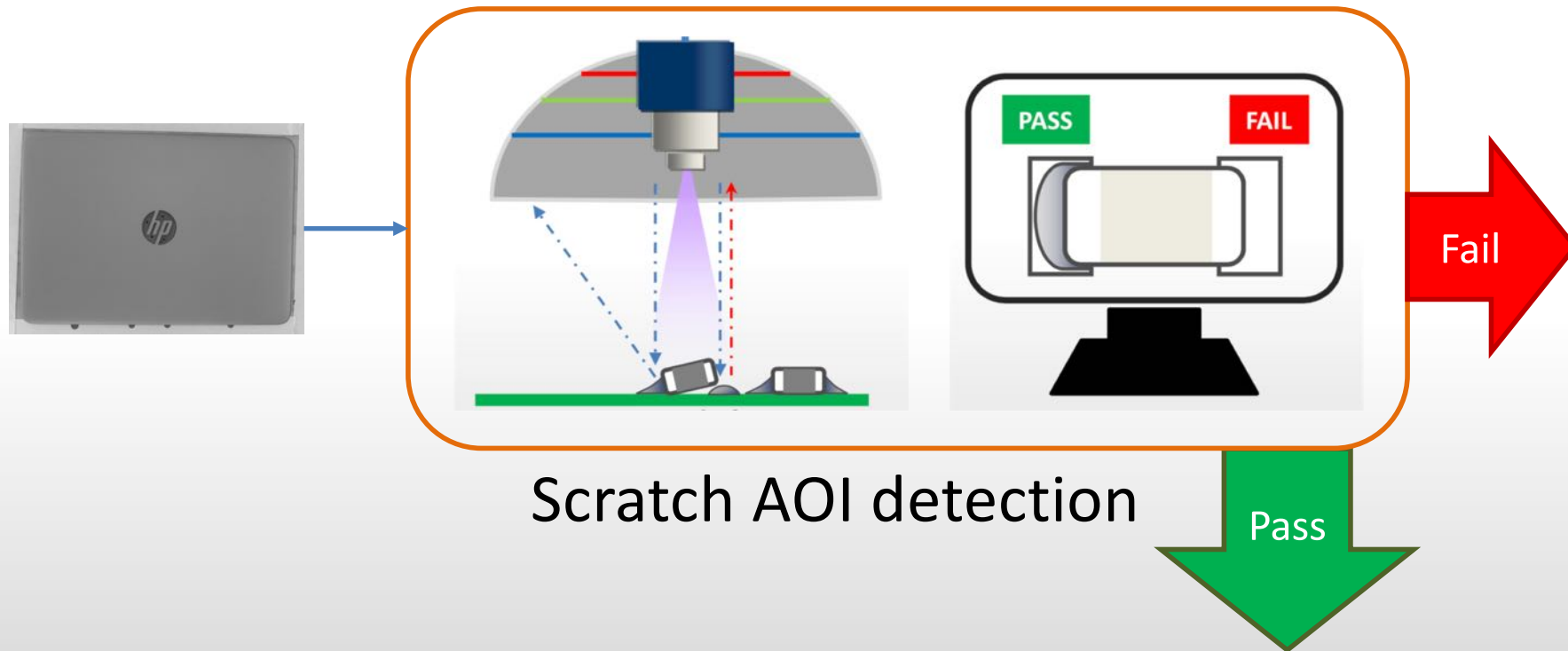
Insufficient solder:

Overflow solder:

(+) Low baseline accuracy without reinspection due to high false positive from first-stage

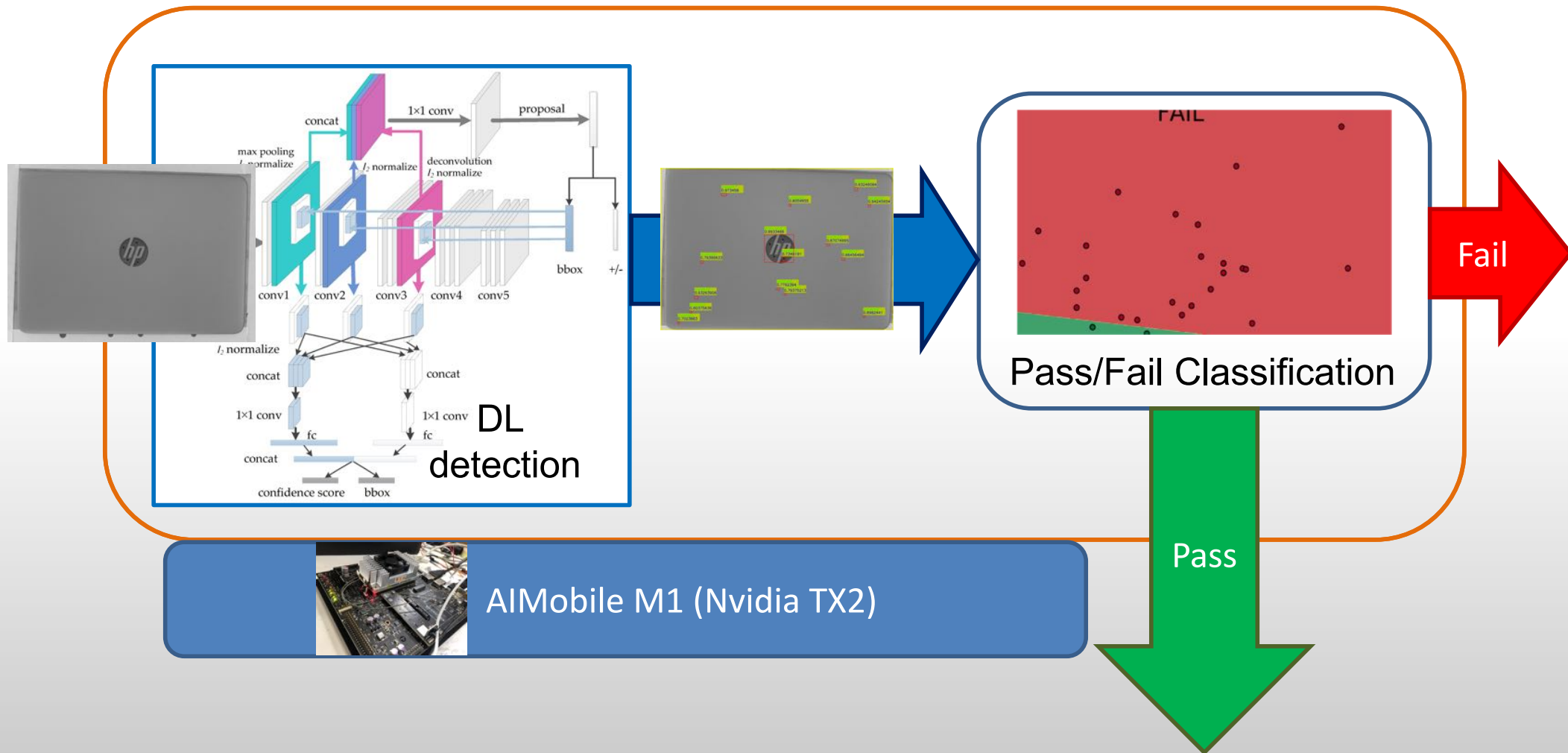
(*) Overall accuracy depending on the first-stage AOI accuracy

2. Laptop surface defect detection



By: Benson Lin of Skywatch, Trista Chen of Inventec, Wei-chao Chen of Skywatch & Inventec

2. Laptop surface defect detection



Edge AI in Smart Manufacturing: Defect Detection and Beyond

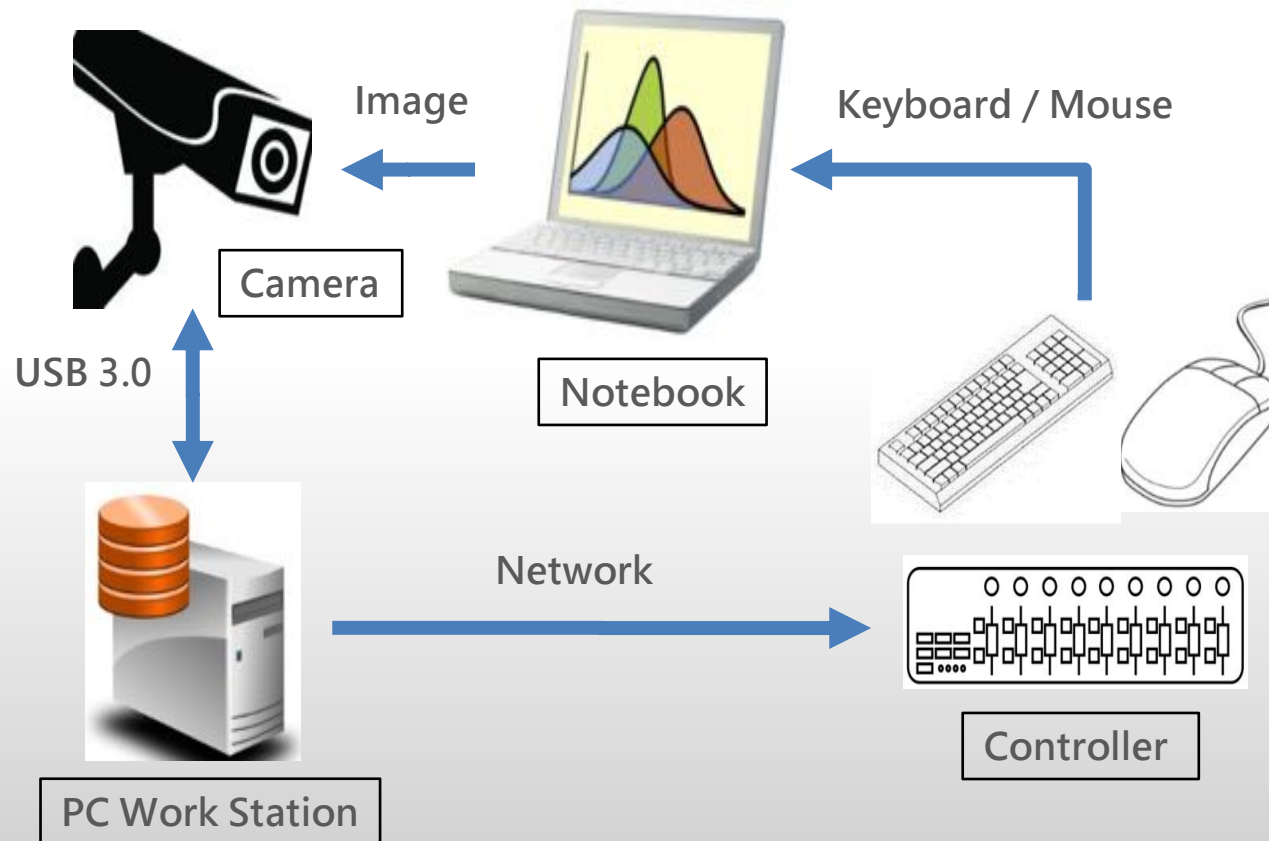
Supplementary material
for 2019 Nvidia GTC talk proposal

by Benson Lin, Skywatch Inc.

Trista Chen, Inventec Corporation

Wei-chao Chen, Skywatch Inc. & Inventec Corporation

3. Smart QA & Testing – Automatic **LA**p**TO**p **TE**sting Robot - LATTE



By: Skywatch (Jimmy Ou, Wei-chao Chen, et al.)

Automatic **LA**p**TO**p **TE**sting Robot - LATTE

❑ Automated control :

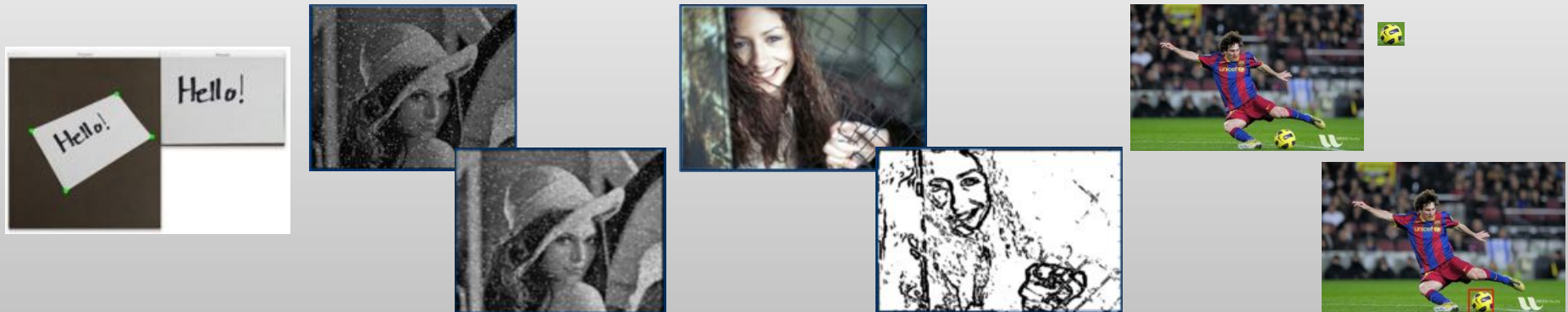
- ✓ USB HID Keyboard Mouse Joystick composite device
- ✓ Pressing device for internal keyboard

❑ OpenCV :

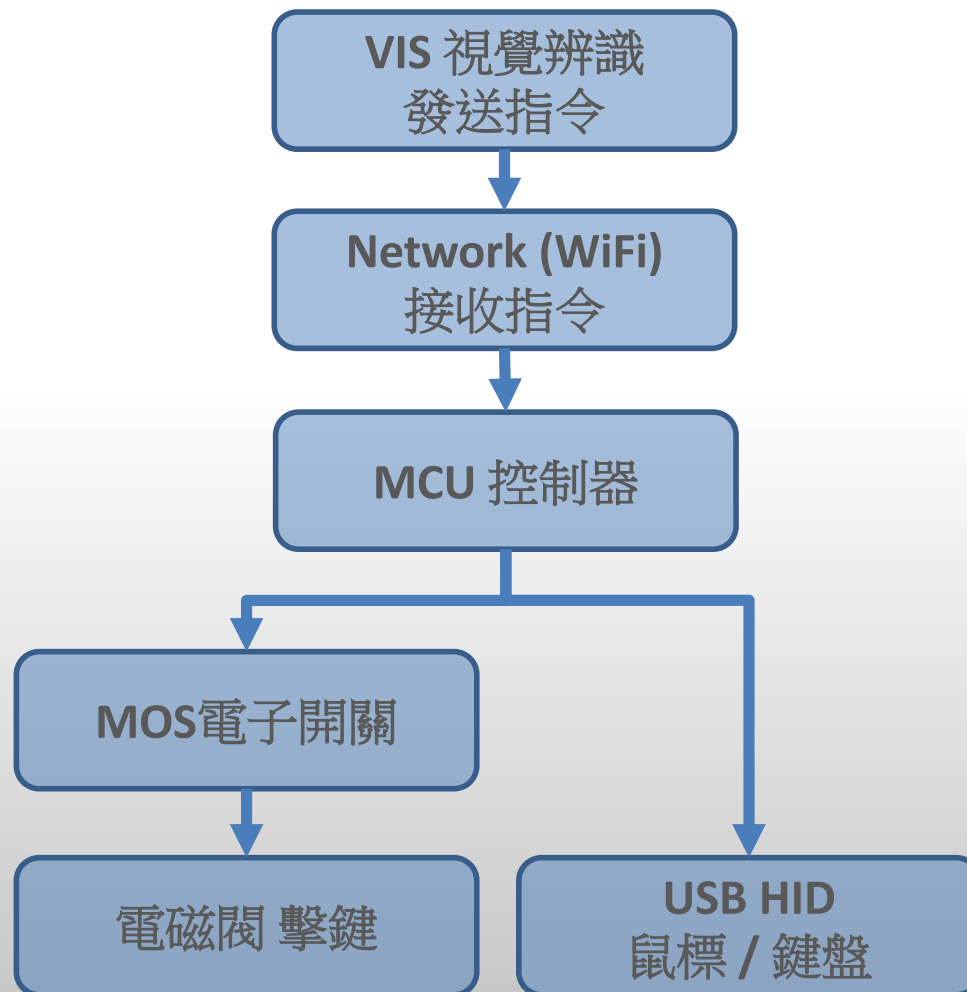
- ✓ Perspective transform
- ✓ Edge detection and denoising
- ✓ Template matching

❑ Tesseract OCR :

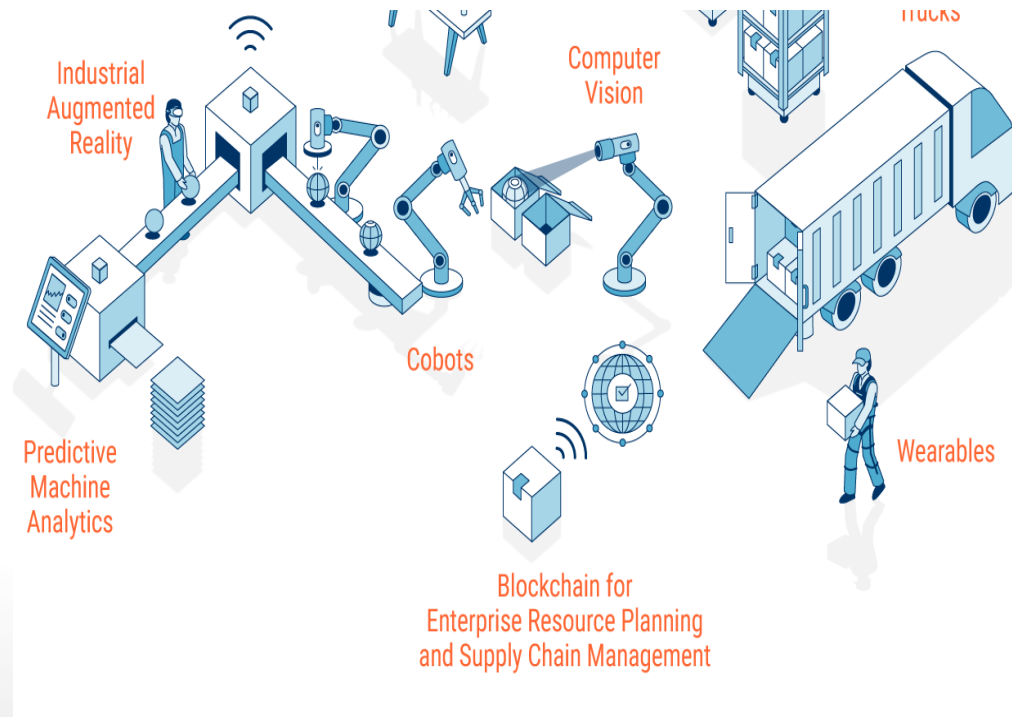
- ✓ LSTM neural networks based OCR engine



Automatic **LA**p**Top** **TE**sting Robot - LATTE







AI • Manufacturing

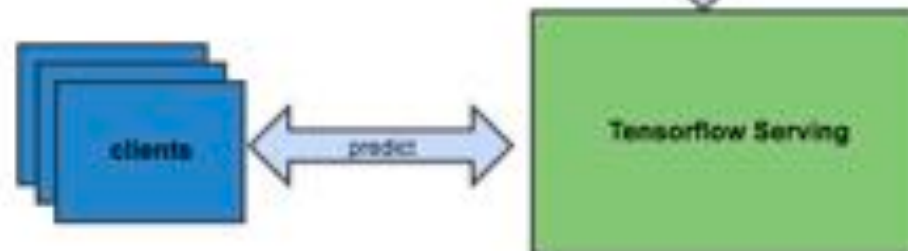
Adopting AI for Manufacturing

Deployable AI

DL “Training”

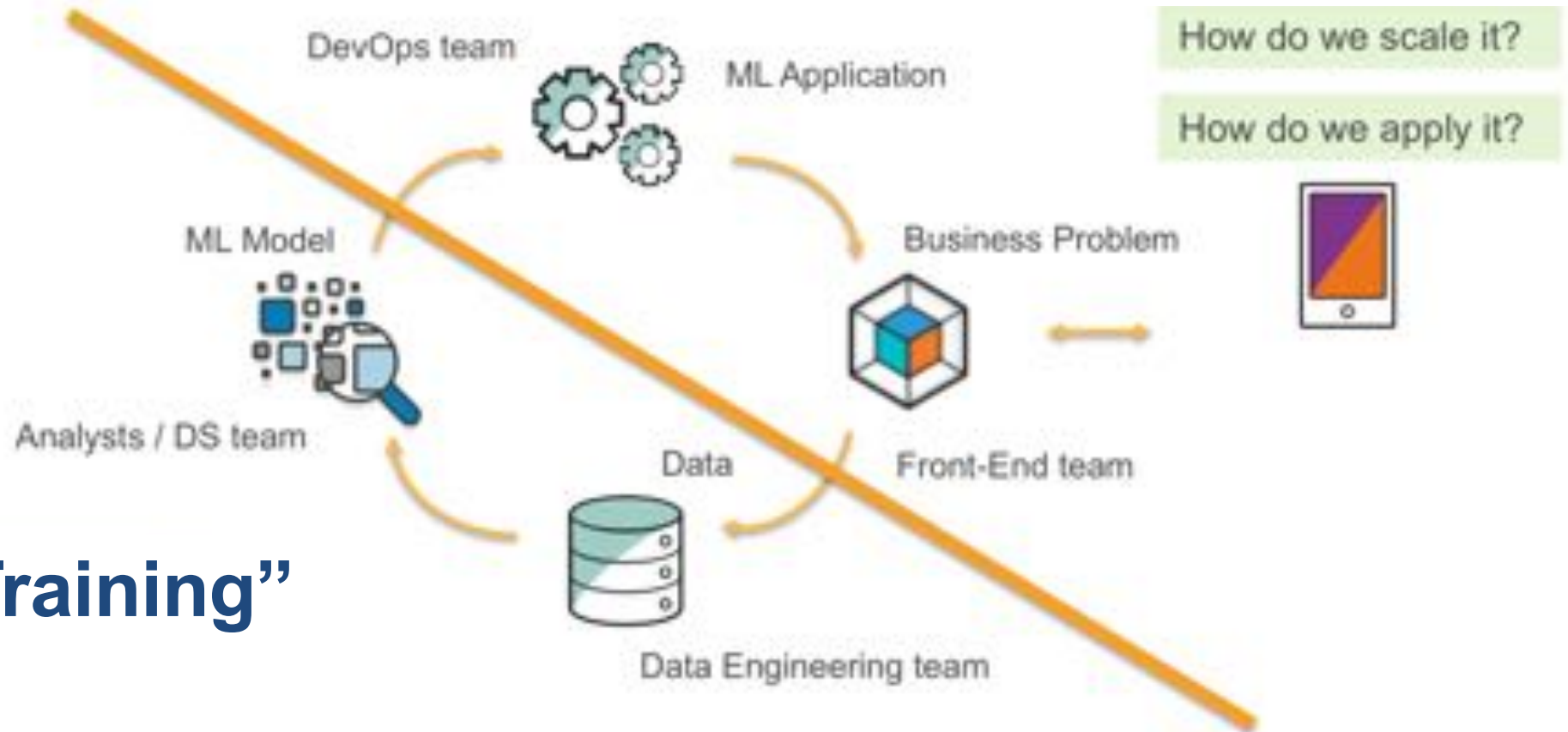


DL “Serving”



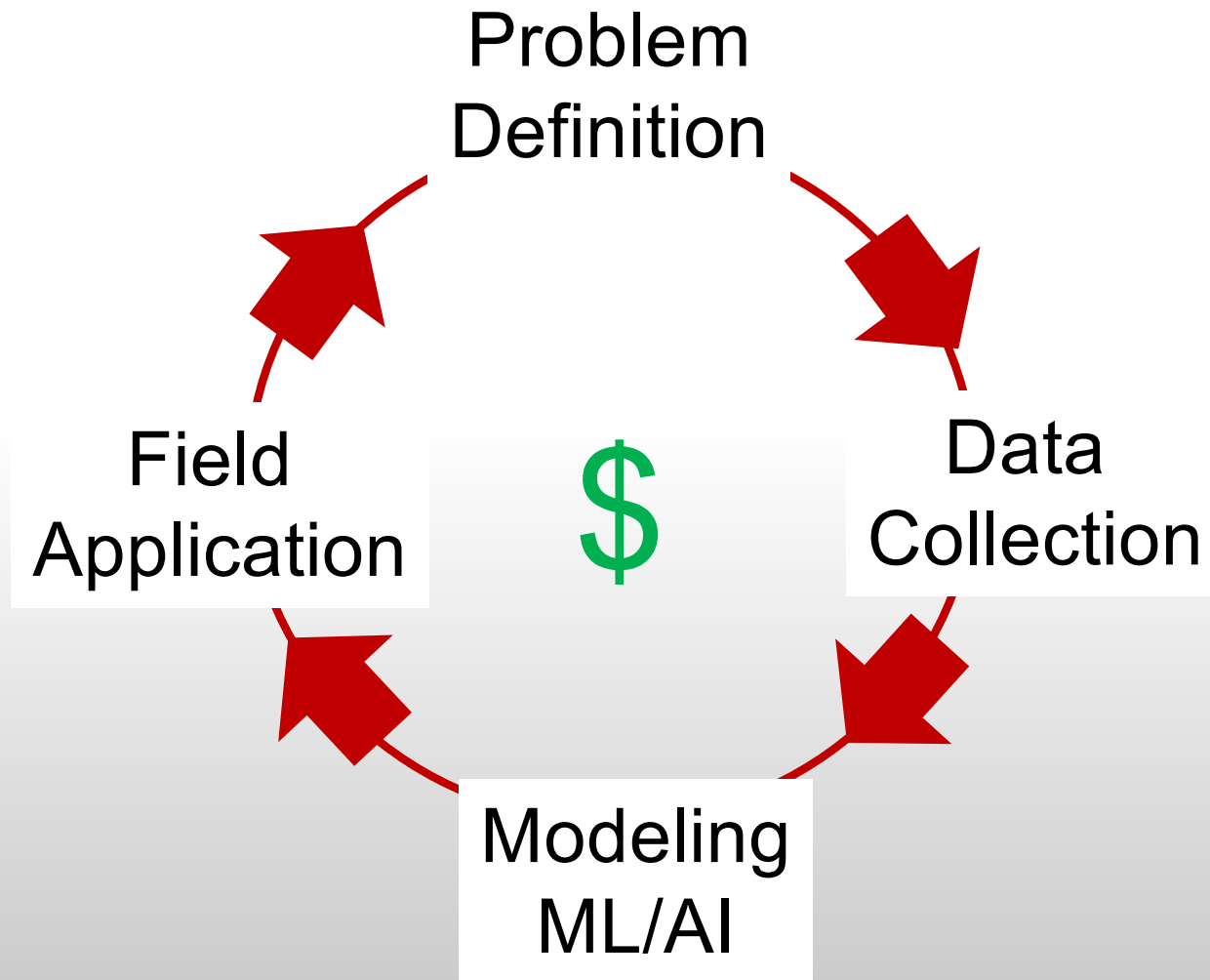
Deployable AI: DL “Training” & DL “Serving”

DL “Serving”

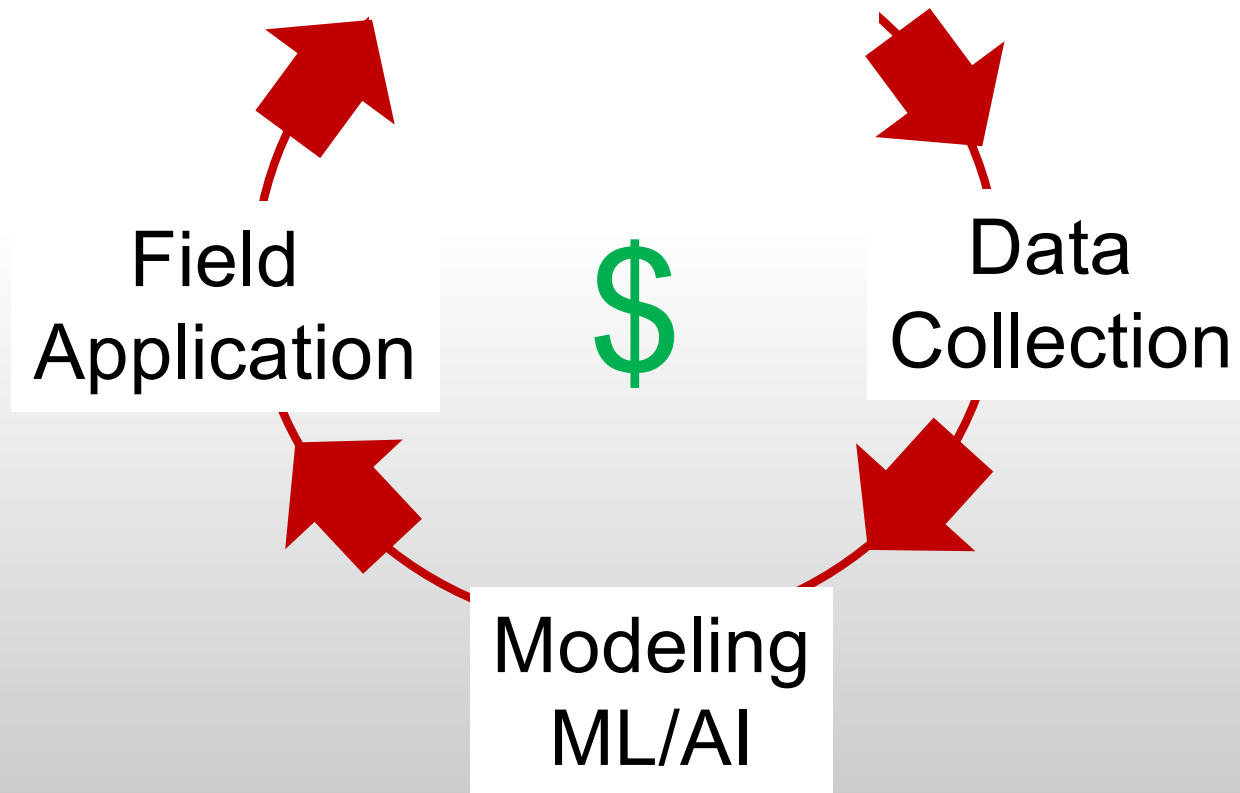


DL “Training”

Deployable AI: DL “Training”



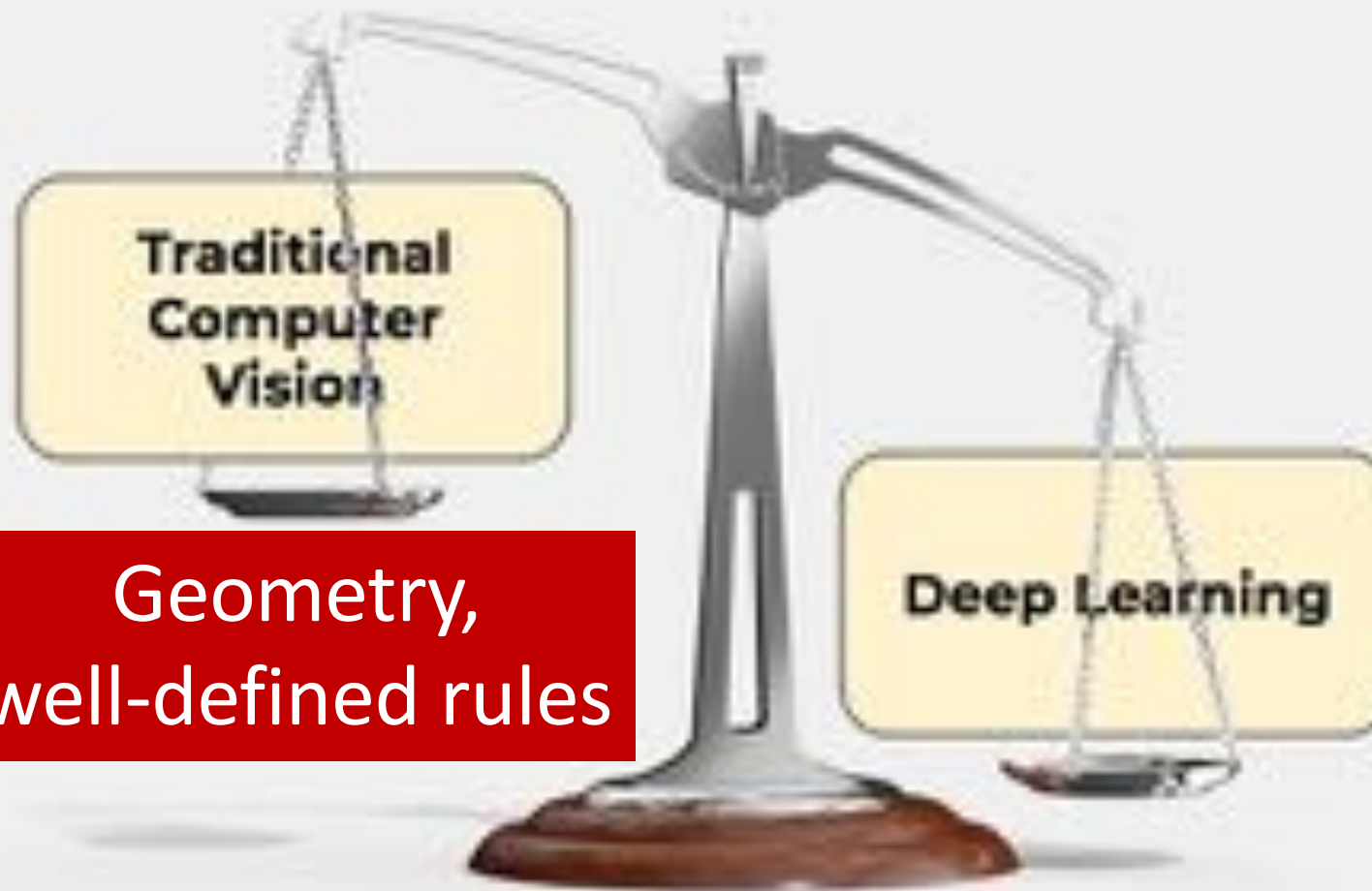
Problem Definition



「我想了一下我平常的工作內容，發現我好像一直在講話，」陳佩君笑說，「寫演算法是最後的工作，對我們來說，溝通反而是最重要的。」在英業達裡，AI中心有兩個面向的任務，一是思考各BU的產品如何加入AI，另一則是如何透過AI，優化企業內的作業流程。

Source: Trista Chen, Business Weekly, Sept 2018

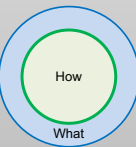
Problem Definition: Not all problems are equal



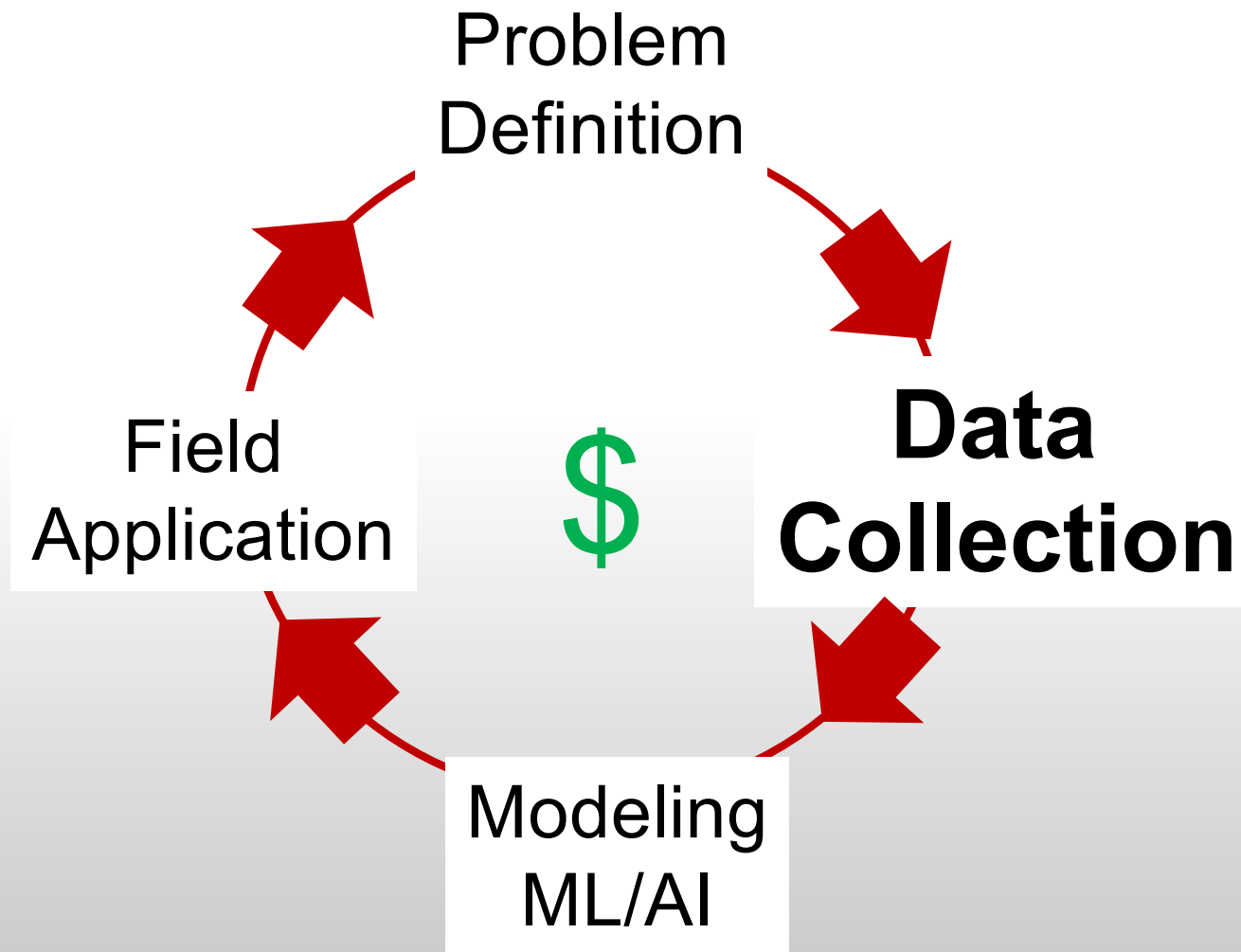
Cognitive TV queries, e.g. 4 Million TVs sending 864k queries/day = 3.5 trillions/day

Google searches: 2.8 Billion searches/day (year 2015)

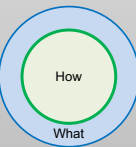
1000 Times more queries than Google!



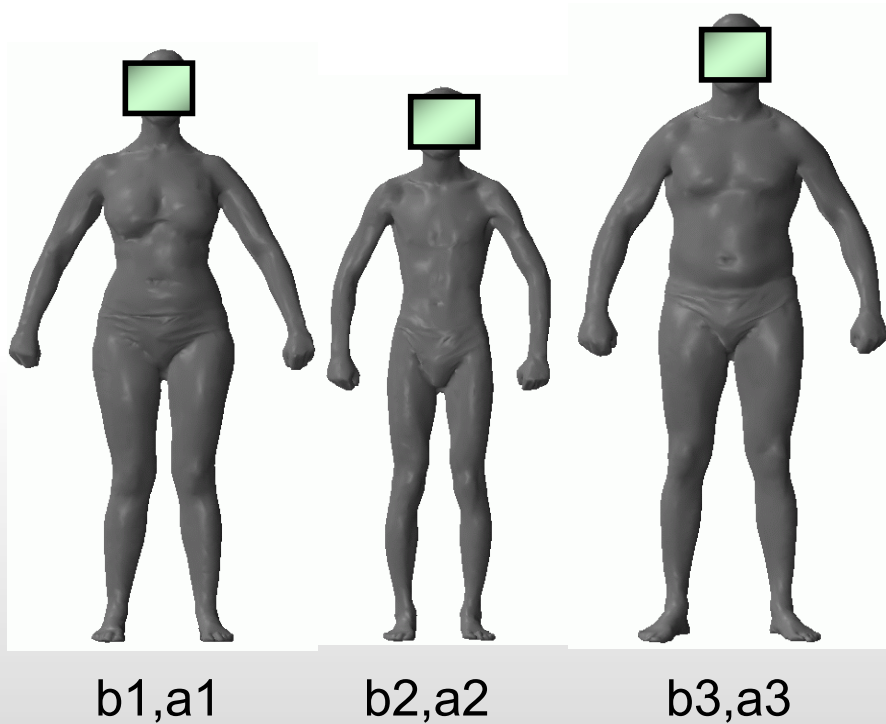




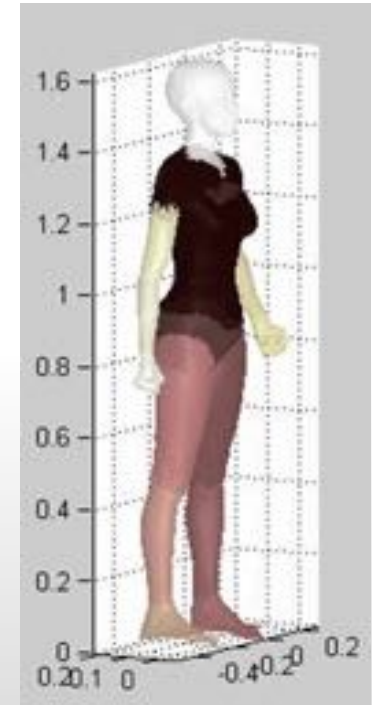
Better features: Intel Body Tracking



Better features: Intel Body Tracking

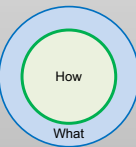


$$a^{\text{new}} \rightarrow b = M * a^{\text{new}} \rightarrow$$



Predict shape coefficients from anthropometric measurements using regression

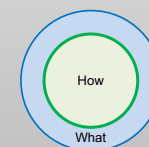
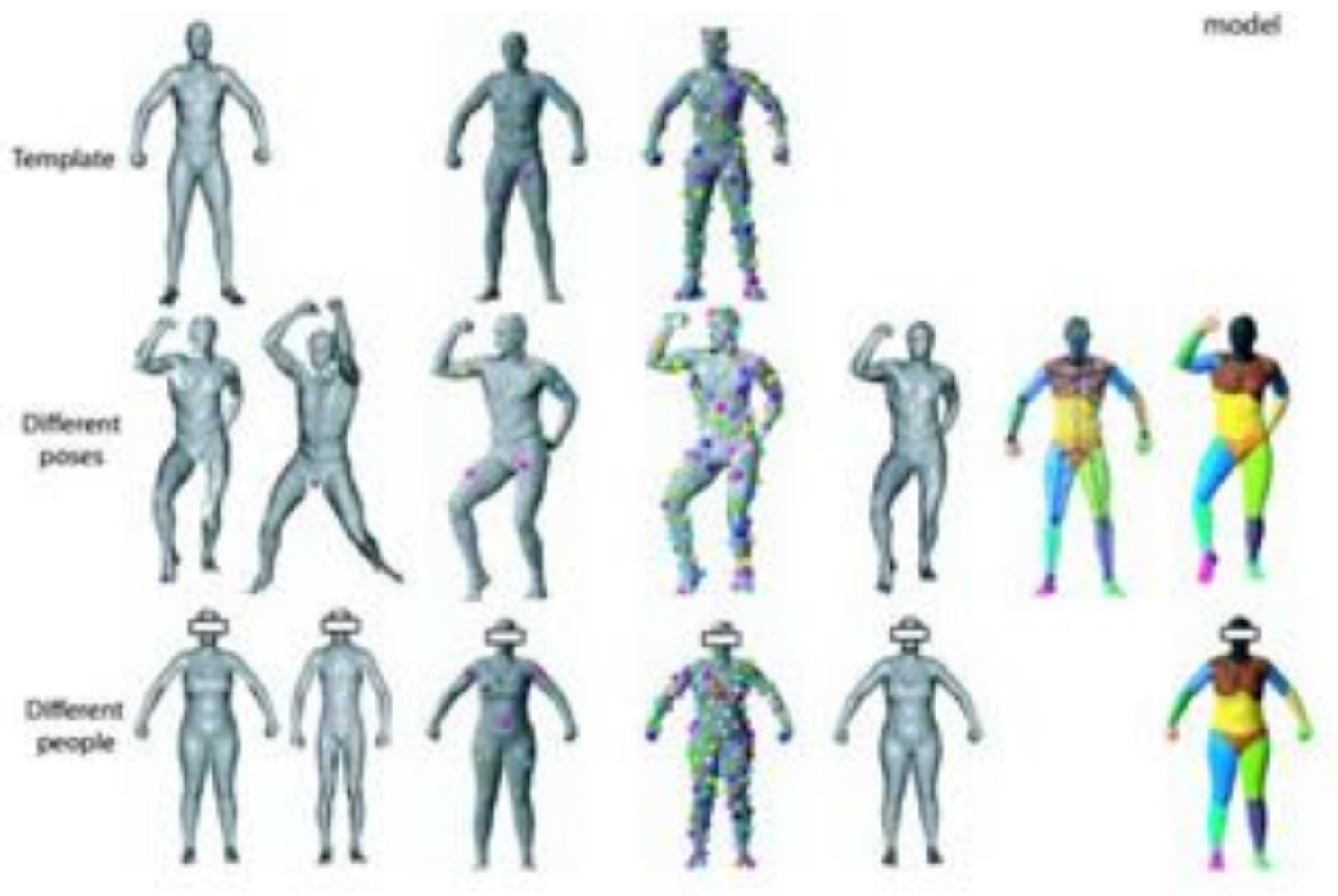
$$[B] = M * [A ; 1]$$



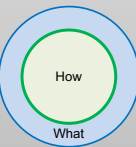
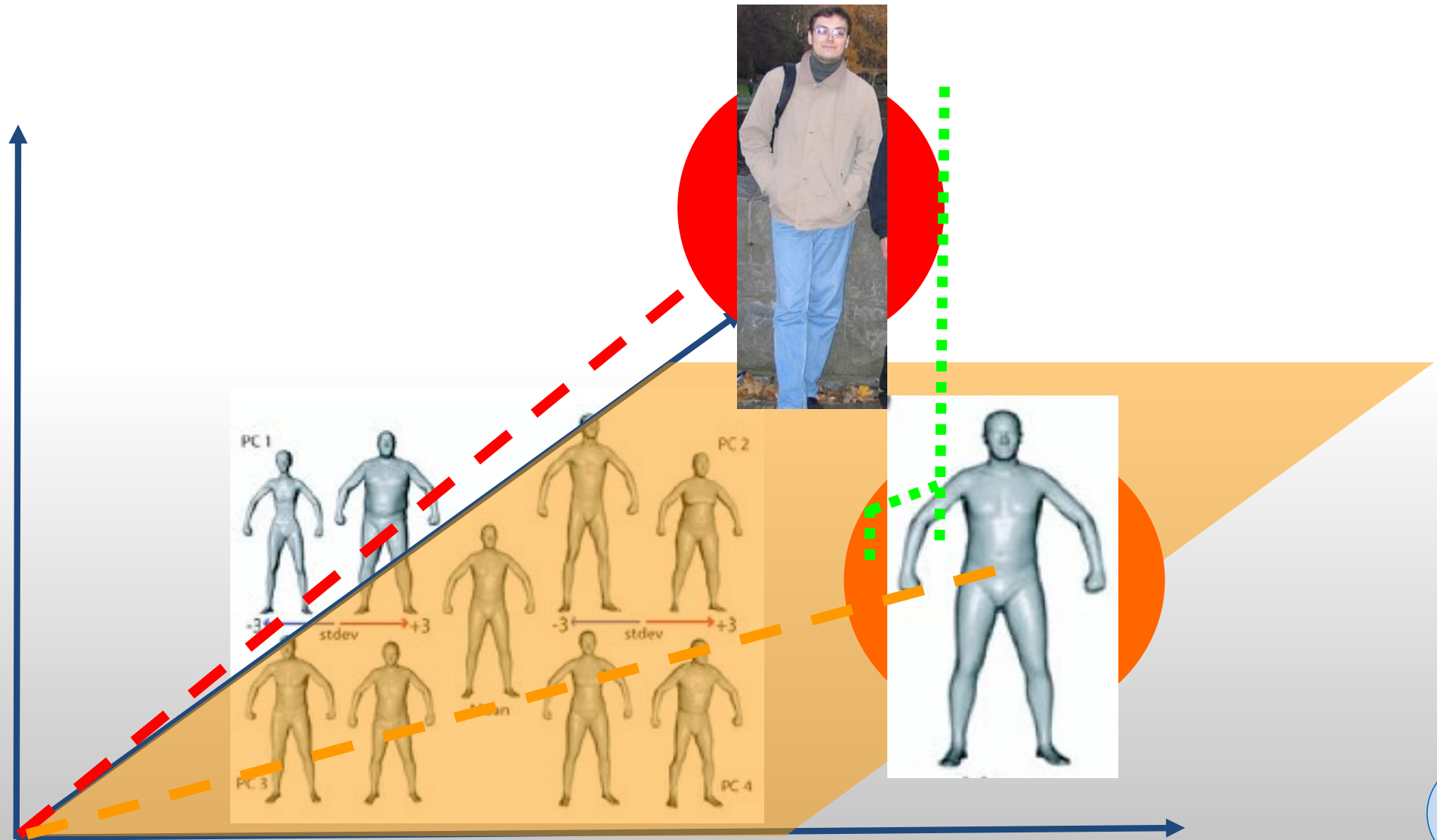
Better features: Intel Body Tracking

SCAPE: PCA of 3D meshes on Poses and Shapes

[Anguelov et al 2005] SCAPE: Shape Completion and Animation of People



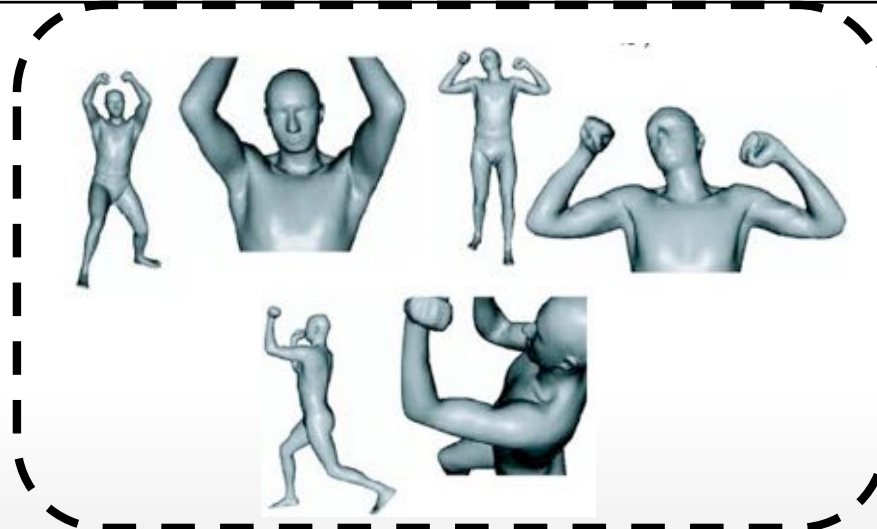
POCS: step 1: Shape deformation



POCS step 2: Pose Deformation

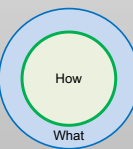
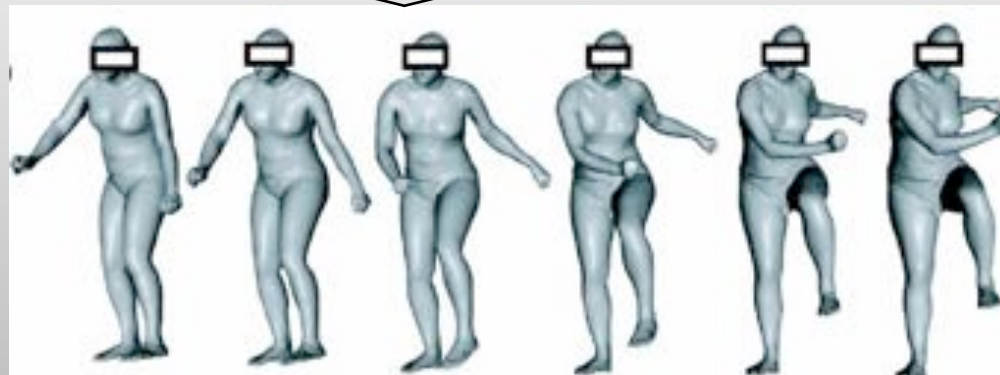


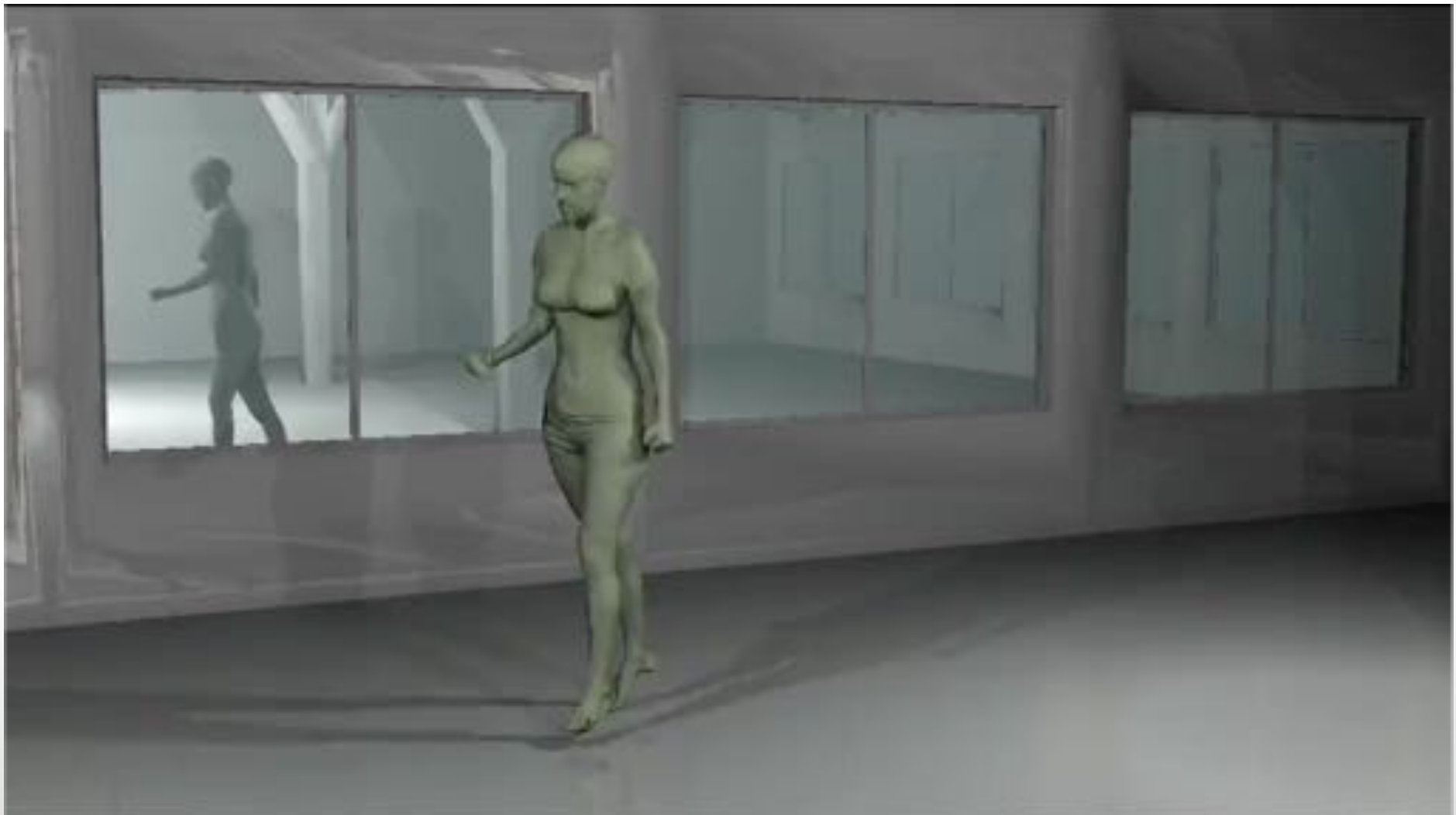
**Data from
Body
Tracker**

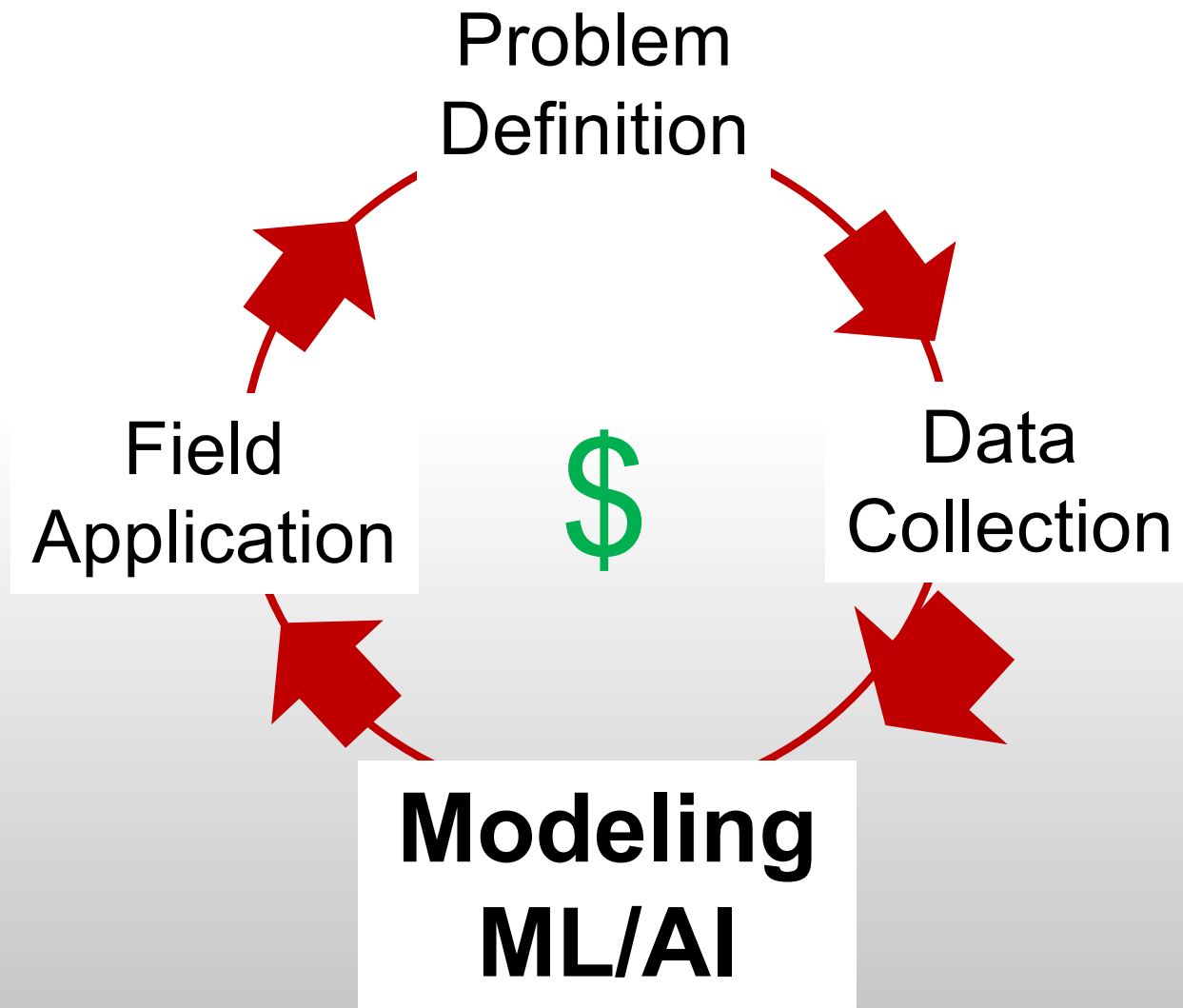


**Pose
Deformation
function**

**Animated
3D model**







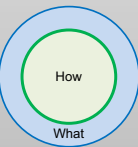


Model selection



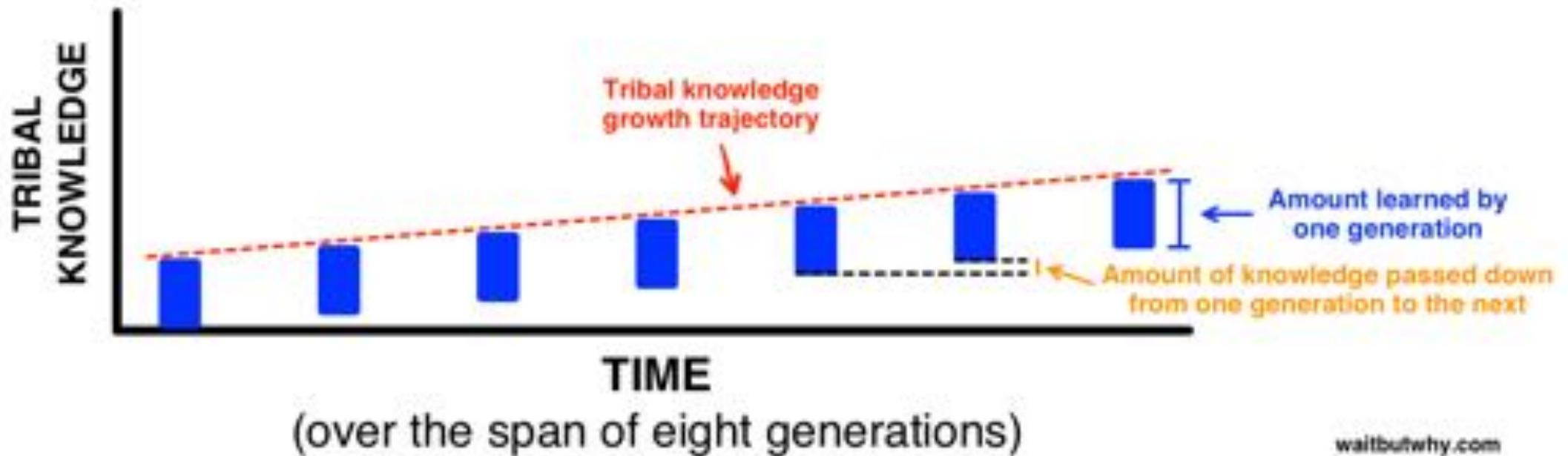
Humans grasp nicely in about a few months (700 wake hours)

- Learning-based method takes as long, supervised learning
- Taught by human pilots: 60s% accuracy



Transfer Learning?

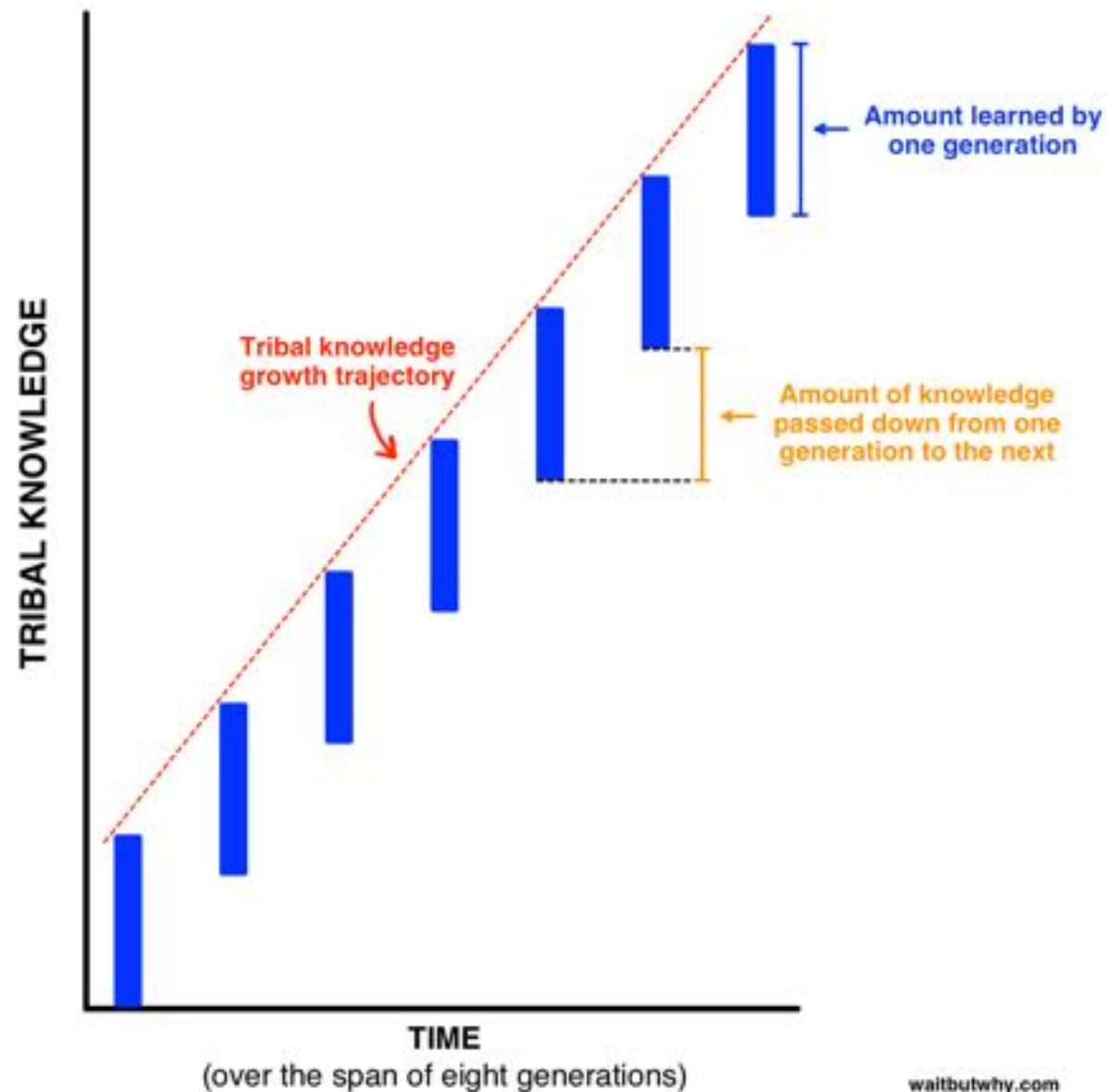
Tribal Knowledge Growth Before Language



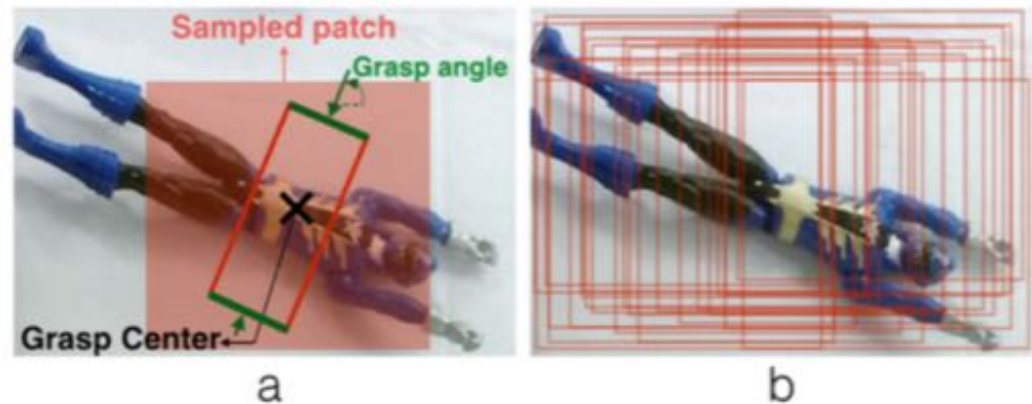
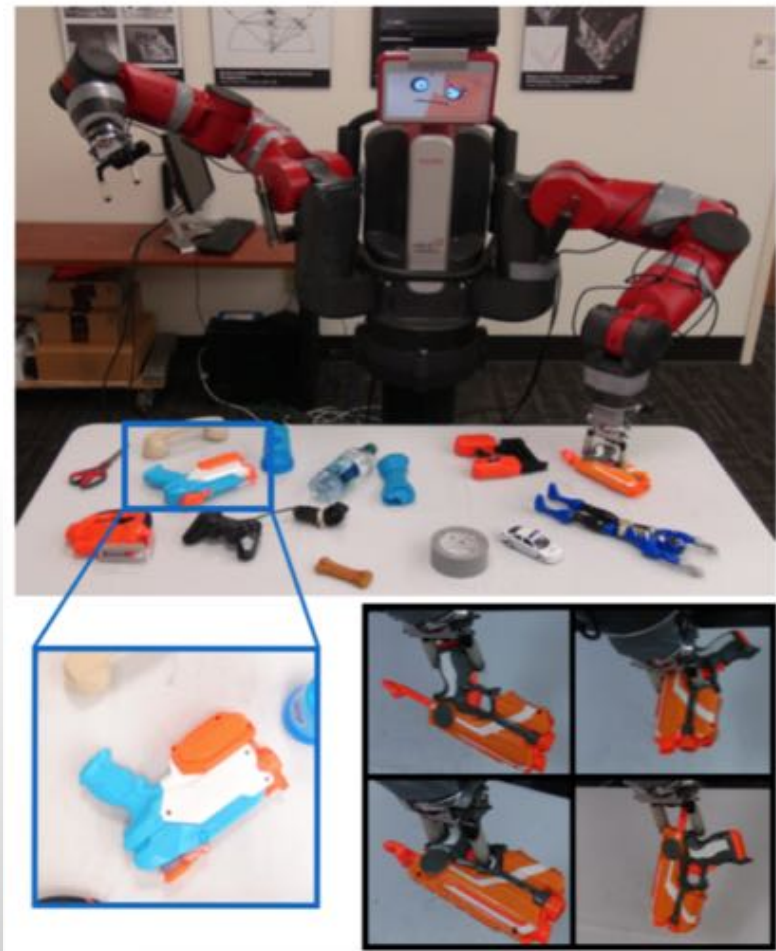
waitbutwhy.com

<https://www.analyticsvidhya.com/blog/2017/06/transfer-learning-the-art-of-fine-tuning-a-pre-trained-model/>

Tribal Knowledge Growth After Language



Transfer learning: image to physical



Domain transfer:

- Right patch $\rightarrow$ center of gravity
- 80% accuracy

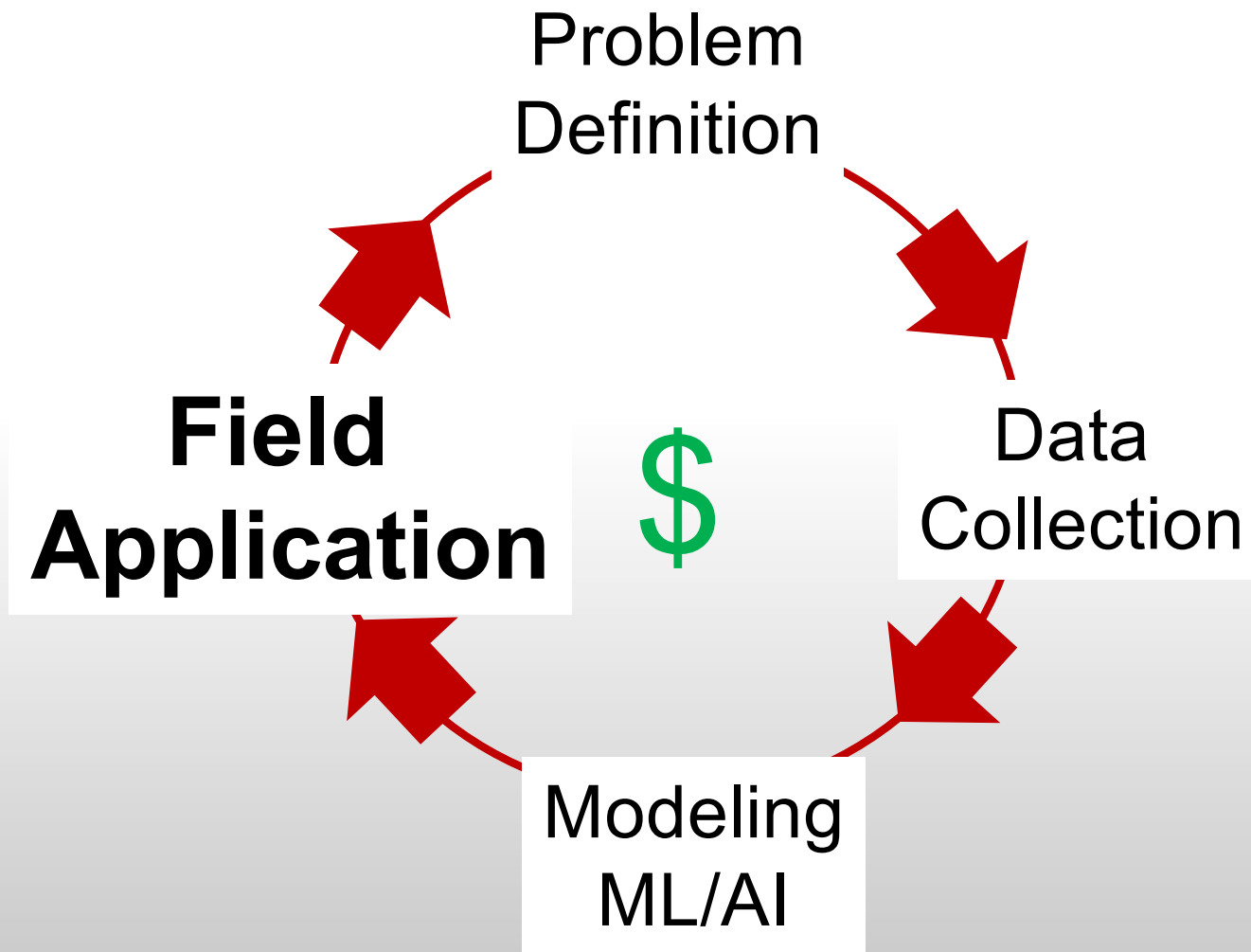
Source: Lerrel Pinto, Abhinav Gupta, "Supersizing Self-Supervision: Learning to Grasp from 50K Tries and 700 Robot Hours", 2015

Reinforcement learning

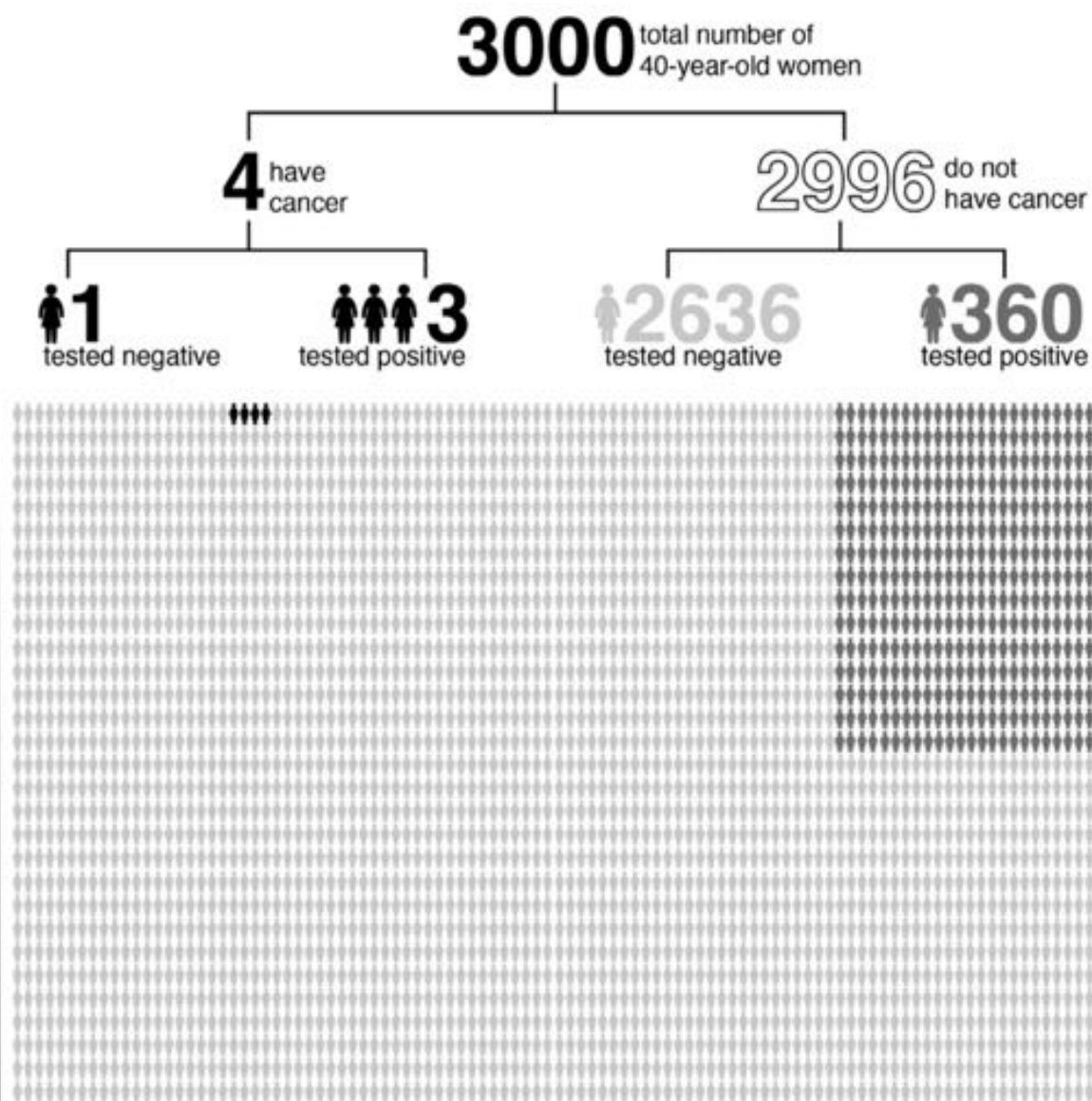


- Actively probes, repositions, handles disturbances
- 96% accuracy

Source: Dmitry Kalashnikov, Alex Irpan, Peter Pastor, Julian Ibarz, Alexander Herzog, Eric Jang, Deirdre Quillen, Ethan Holly, Mrinal Kalakrishnan, Vincent Vanhoucke, Sergey Levine
“QT-Opt: Scalable Deep Reinforcement Learning for Vision-Based Robotic Manipulation”,
<https://arxiv.org/abs/1806.10293>, June 2018



Field Application: Metrics



Test\Truth	Cancer	No Cancer
Test Positive	3	360
Test Negative	1	2636

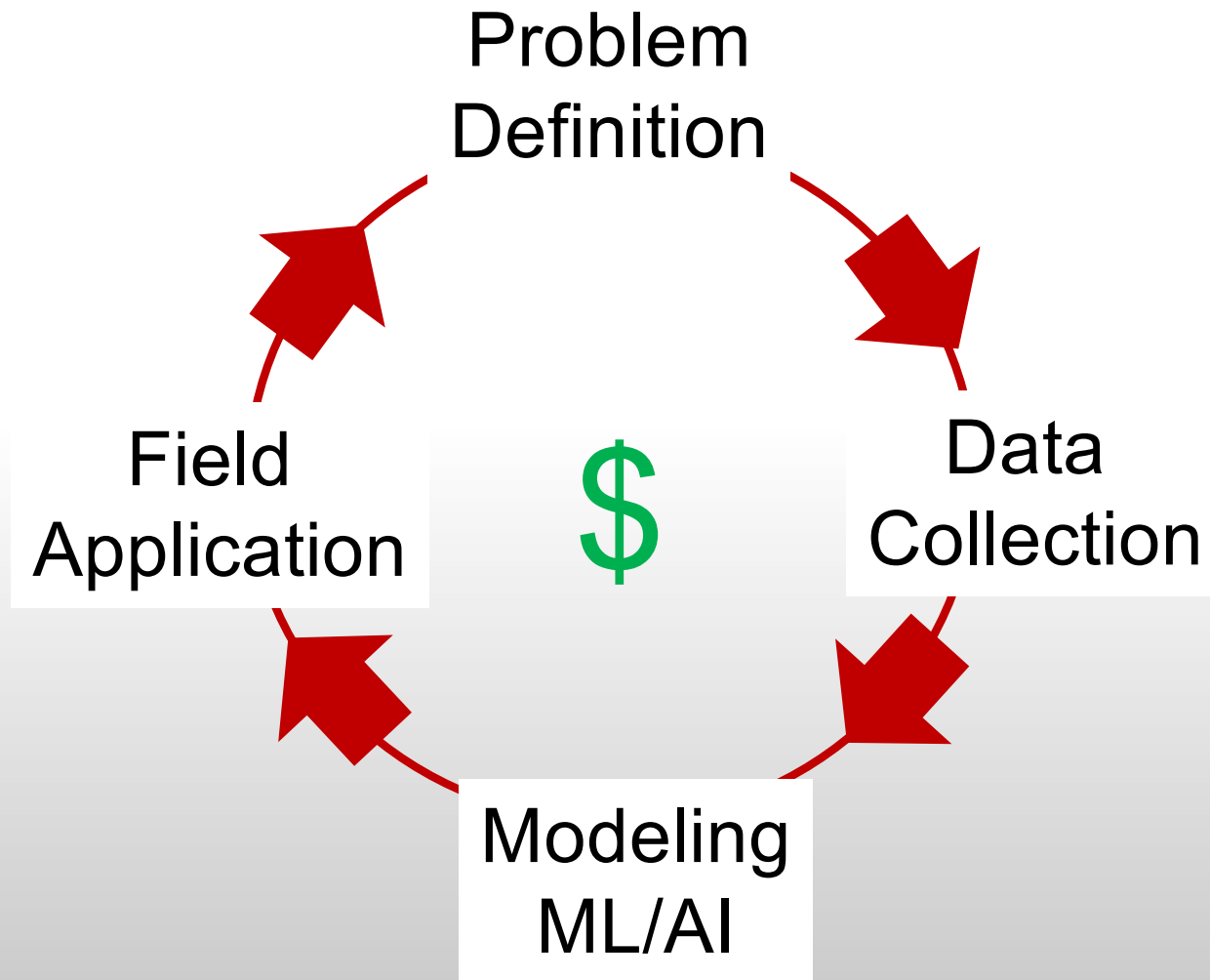
Accuracy = 88%

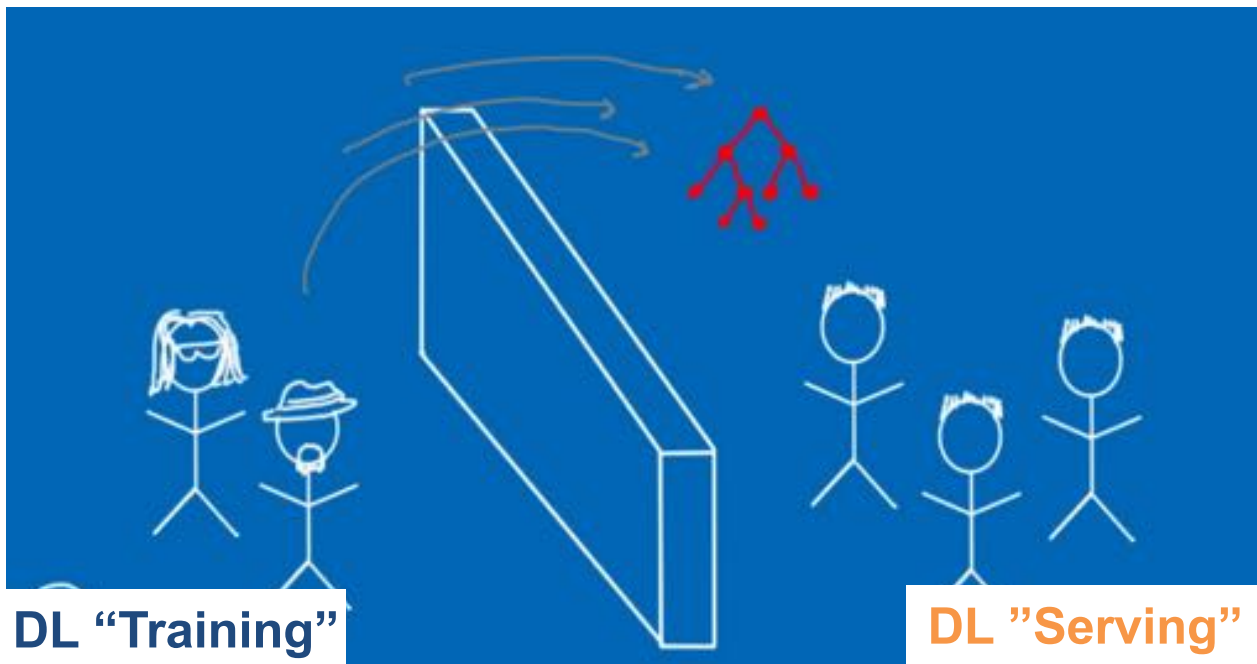
Recall = $\frac{3}{4} = 75\%$

Precision = 0.8%

Source: Judea Perl,
“The Book of Why”, 2018

Deployable AI: DL “Training”

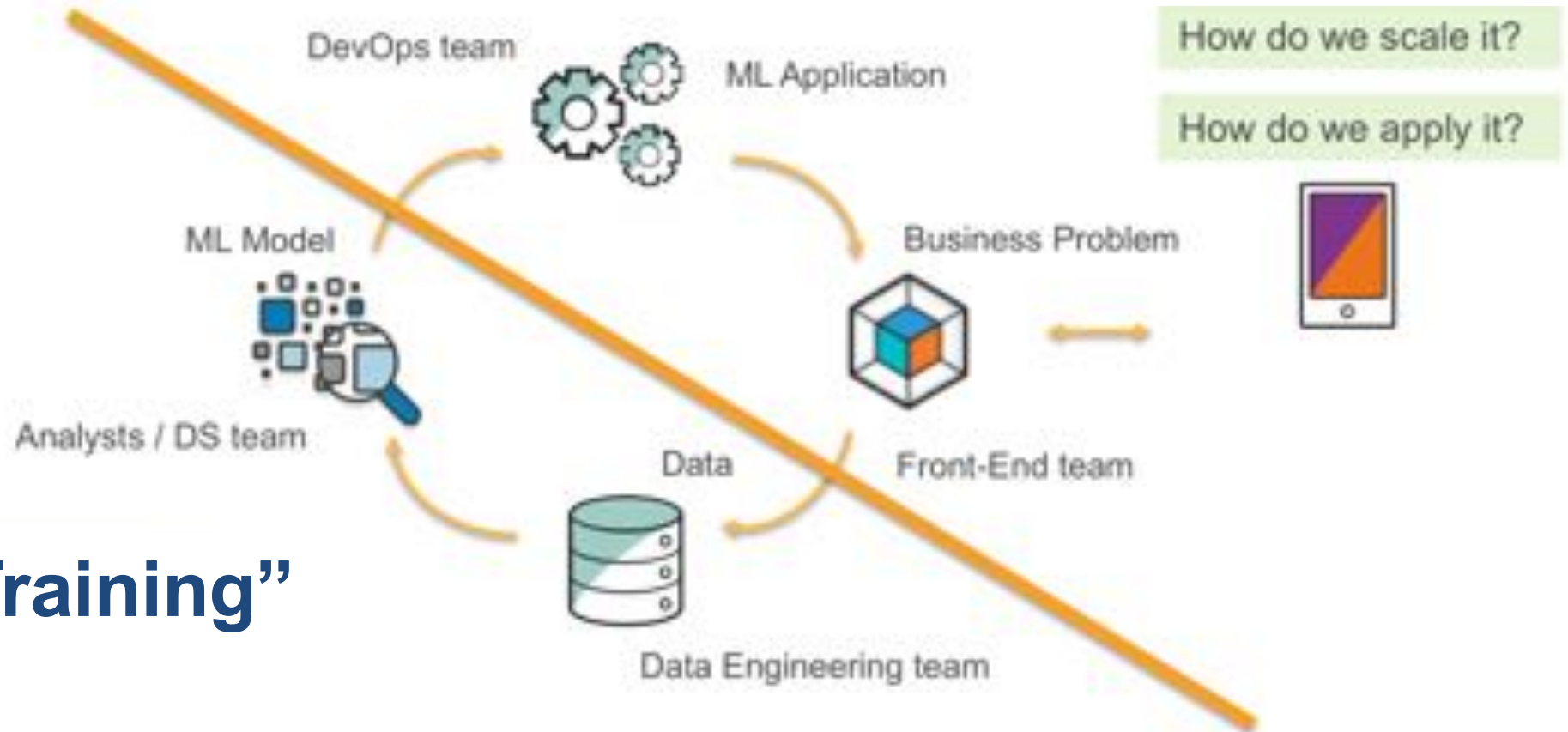




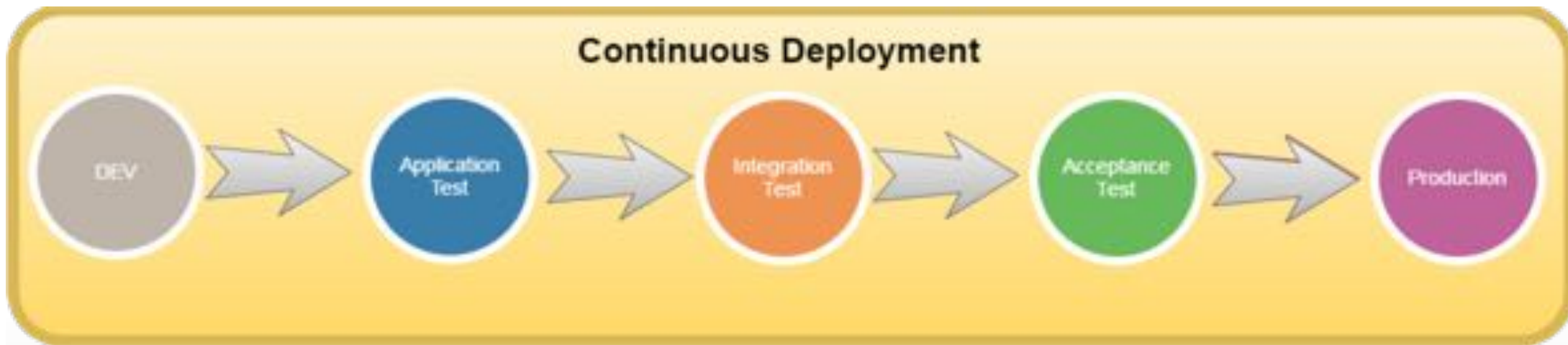
Deployable AI: DL “Training” & DL “Serving”

DL “Serving”

DL “Training”



Idea: Continuous Integration and Deployment

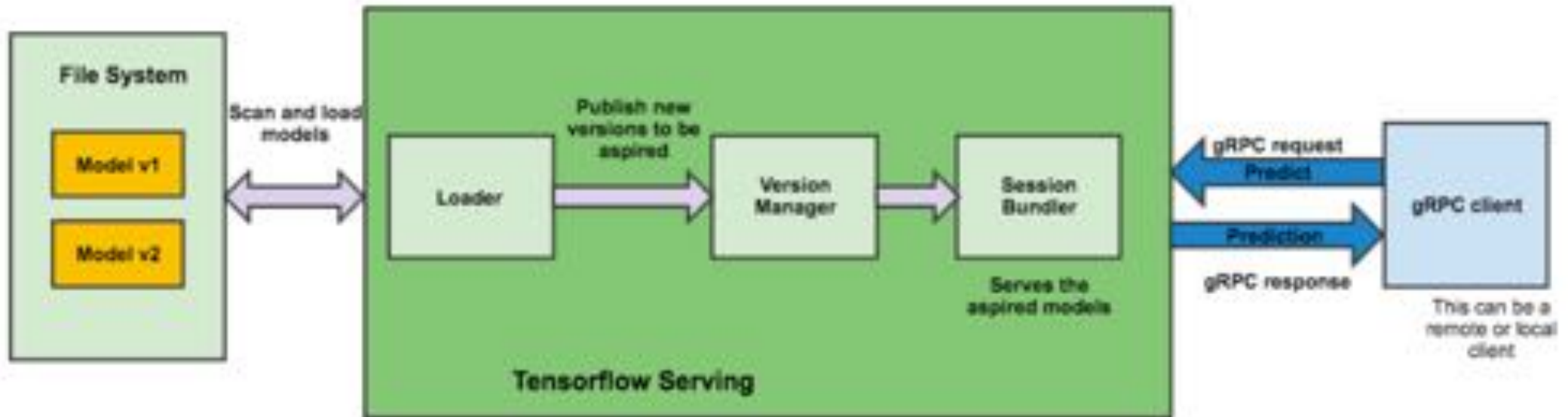


DL Training
(The “**Code**” with its
Parameters/Weights)

DL Serving
DevOps (The “**Service**”)

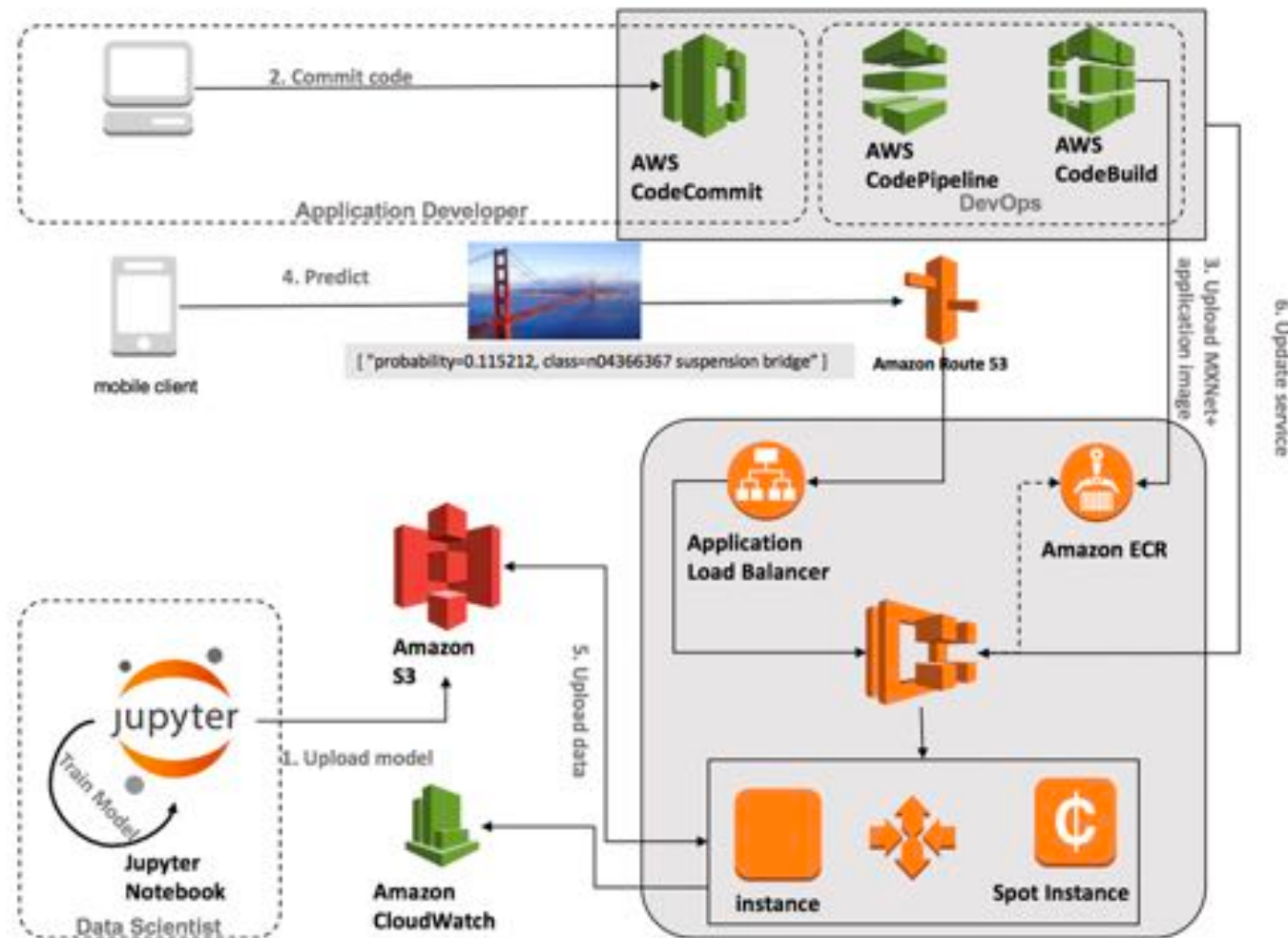
Deployable AI: DL “Serving”

Example #1: Google TensorFlow Serving



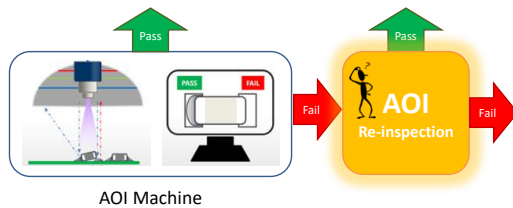
Deployable AI: DL “Serving”

Example #2: AWS ECS



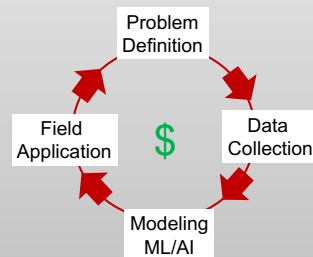
Summary

- AI for Smart Manufacturing Case Studies



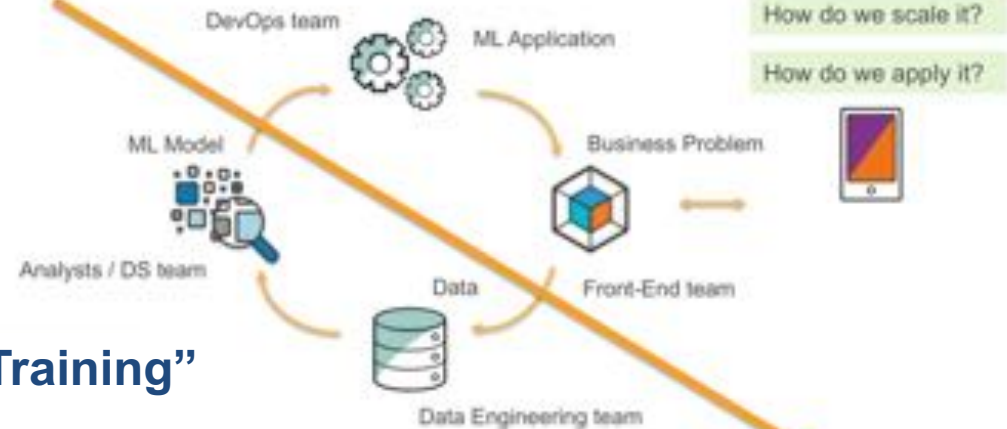
- Deployable AI

1. DL training
2. DL serving



DL “Training”

DL “Serving”





THANKS !!!

Questions? Joining? Email: chen.trista@inventec.com